

$\mathcal{O}(n)$ alternative to Quantum Fourier Transform with efficient neural net classical post-processing

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The Quantum Fourier Transform (QFT) is required by hidden subgroup problem (HSP) algorithms, including Shor’s algorithm for factoring. The circuit depth of the QFT remains challenging for near-term hardware. To find shallower alternatives we identify two properties that are exploited by the QFT to enable HSP. Firstly, the shift invariance of the QFT allows for the removal of a random overall shift. Secondly, the QFT retains information about the hidden subgroup generator accessible in the measurement outcomes. We quantify that information via the discrete Fisher information. We construct a family of shallow circuits using Hadamards and controlled-Phase gates, HP- L circuits, that we prove preserve shift invariance. Numerical analysis shows these circuits retain exponentially growing Fisher information. The $\mathcal{O}(n)$ HP-1 can replace the $\mathcal{O}(n^2)$ QFT in Shor’s algorithm, as demonstrated numerically, with an efficient neural network implementing classical post-processing.

Introduction.—The $\mathcal{O}(n^2)$ circuit depth required by the standard Quantum Fourier Transform (QFT) presents experimental challenges. The QFT is used in the exponential quantum speedups of landmark protocols—such as Shor’s algorithms for prime factorization and discrete logarithms [? ? ?]. The associated family of algorithms, all using the QFT, is known as the (quantum) hidden subgroup problem (HSP) algorithms [? ? ? ?]. The experimental implementation of the QFT remains challenging. For a 100-qubit system ($n = 100$), the QFT demands thousands of circuit layers, which vastly exceeds current hardware capabilities. For instance, IBM’s 127-qubit Eagle processor is limited to a depth of 60 layers even with extensive error mitigation [?]; Google’s 105-qubit Willow processor reaches its operational threshold at around 40 layers [?]; USTC’s 105-qubit Zuchongzhi 3.0 processor demonstrated random circuit sampling on 83-qubits, with around 32 layers [?]. Moreover, for quantum error-mitigation protocols that are commonly used, the sampling overhead can scale exponentially with the circuit depth [? ?].

There has been significant progress in reducing the circuit depth required. These efforts have enabled record experimental QFT fidelities [?] through techniques including (i) approximate or truncated QFT constructions, which omit small-angle controlled rotations [? ?] with $\mathcal{O}(n \ln n)$ depth, (ii) dynamic mid-circuit measurements, which reduce the resource scaling of QFT followed immediately by measurement from $\mathcal{O}(n^2)$ two-qubit gates to $\mathcal{O}(n)$ mid-circuit measurements without connectivity constraints [?], and (iii) quantum circuit cutting and cluster-simulation methods, which trade coherent quantum depth for additional sampling and classical post-processing overhead [? ?].

We here take a different approach to reducing the circuit depth required. Rather than approximate the QFT we identify key features of the QFT that make it compatible with the HSP. We focus on shift-invariance combined with retention of information about the generator in question. Fig. ??(a) summarizes the main idea. We design non-QFT circuits that retain those key features.

We construct a circuit family, called HP- L circuits, that preserves shift invariance. In particular, the HP-1 circuits are shallow and retain information about the generator. The HP- L circuits interleave transversal Hadamard (H) gates with L layers of phase (P) blocks. We prove analytically that HP- L circuits are shift-invariant. Numerical scaling analysis demonstrates exponentially increasing retention of Fisher information $\sim e^{kn}$, albeit with a worse exponent k than QFT. That $\sim e^{kn}$ -scaling consistent with efficient scaling of the number of samples required.

We numerically simulate replacing the QFT in Shor’s algorithm with HP-1 circuits, and using an efficiently scaling neural network [?] for classical post-processing. Factorisation is successfully achieved.

Necessary condition for Shift Invariance.—The standard quantum HSP algorithm is illustrated in Fig. ??(b) with gray blocks [? ? ? ?]. After U_f , the state is proportional to $\sum_x |x\rangle |f(x)\rangle$. The elements x belong to a finite abelian group G with group operation $+$. We are guaranteed that $f(x+r) = f(x)$ for the unknown generator r that we wish to find. In other words, the function $f(\cdot)$ is constant on cosets of V , the subgroup generated by r . Upon measuring (or imagining measuring) the second register to find $|f(c)\rangle$ the state becomes proportional to $\sum_{q \in \{0,1,2,\dots\}} |c+qr\rangle |f(c)\rangle$, where the shift c acts as a random coset offset and does not af-

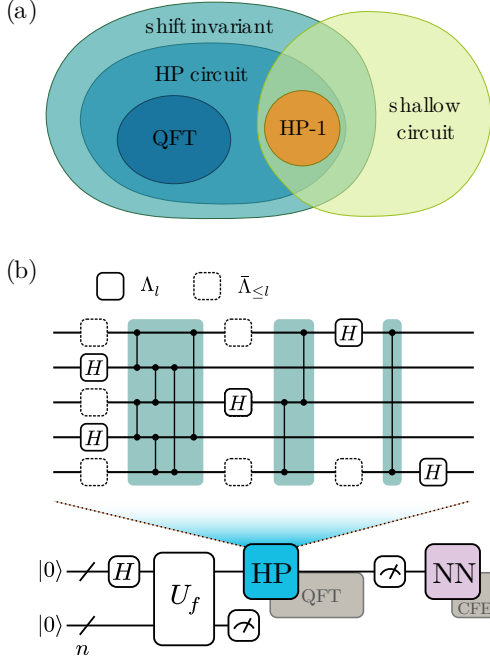


FIG. 1. **Shallow shift-invariant alternative to the QFT.** (a) Shift invariance is a critical property of the QFT. We design a circuit family, denoted as HP- L , that preserves the shift invariance. Within this broader family, HP-1 specifically serves as a shallow-depth circuit that is experimentally friendly. (b) In the proposed HSP algorithmic routine, the deep QFT block (the left gray gate) is directly replaced by the HP circuit. A neural network then processes the measurement outcomes, bypassing the standard continued fraction expansion (CFE) typically used in the standard HSP. The HP circuit is composed of Hadamard and phase gates, with Λ_l referring to the qubits with Hadamards applied in layer l .

fect the recovery of r (additional details about the HSP algorithm are provided in Appendix ??).

We now define shift invariance [?] for HSP sampling and derive a necessary condition that any shift-invariant unitary must satisfy. For a subgroup $V \leq G$, in particular $V = \{|qr\rangle\}$, we denote the uniform superposition over its elements as $|V\rangle := \sum_{q \in \{0,1,2,\dots\}} |qr\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} |v\rangle$, for notational simplicity.

Definition 1 (Shift invariance). *A unitary U acting on the group register is shift-invariant for HSP sampling if*

$$|\langle x|U|V\rangle|^2 = |\langle x|U|c+V\rangle|^2, \quad \forall x, c \in G, \quad \forall V \leq G. \quad (1)$$

To identify shift-invariant circuits beyond the QFT, we first establish a necessary condition for shift invariance.

Proposition 1 (Necessary condition of shift invariance). *Suppose U is a unitary on the group register that satisfies the shift-invariance condition above. Then, for every*

$x, y \in G$ there exist phases $\theta_{xy} \in \mathbb{R}$ such that

$$\langle x|U|y\rangle = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}. \quad (2)$$

Proof sketch of Proposition ??. The proof hinges on analyzing the shift-invariance condition for the trivial subgroup $V = \{0\}$. For $V = \{0\}$, the state $|V\rangle$ is the computational basis state $|0\rangle$, and the shifted state is $|c+V\rangle = |c\rangle$. The requirement (Eq.??) that $|\langle x|U|0\rangle|^2 = |\langle x|U|c\rangle|^2$ for all $x, c \in G$ implies that, for any fixed output x , the magnitude of every matrix element in the x -th row of U must be identical. The special case induces the uniform magnitude $\langle x|U|y\rangle = 1/\sqrt{|G|} e^{i\theta_{xy}}$. The full details of this argument are provided in Appendix ??.

L -phase-block Hadamard-Phase circuits (HP): A shift-invariant circuit family.—Proposition ?? suggests a simple design principle for constructing shift-invariant circuits. The condition $|\langle x|U|y\rangle| = 1/\sqrt{|G|}$ requires flat amplitude magnitudes, which are naturally generated by Hadamard gates. The remaining freedom lies in the relative phases θ_{xy} , which can be adjusted by controlled-phase gates $\text{CP}(\theta) = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + e^{i\theta}|11\rangle\langle 11|$ without changing these magnitudes. This motivates the Hadamard-phase (HP) circuit family we shall now introduce. As illustrated in Fig. ?? (b), the HP- L circuit is structured as an alternating sequence of Hadamard gates and phase blocks. To respect shift invariance, we enforce three rules for HP- L : (i) each qubit is acted on by a Hadamard gate exactly once; (ii) each qubit pair is connected by at most one controlled-phase gate; and (iii) each controlled-phase gate must come together with 2 Hadamards, one before and one after the phase gate. Here, “once” refers to the entire L -layer circuit rather than to a single layer.

To express the allowed circuits mathematically, we define two types of sets of qubits. Let H_i denote the Hadamard gate applied to qubit $i \in \{1, 2, \dots, n\}$. For each layer l , let Λ_l denote the set of qubits on which Hadamard gates are applied in that layer. The qubits that are still allowed in terms of having a Hadamard on them after layer l , i.e. those that have not had a Hadamard on them in or before layer l , are denoted as the set $\bar{\Lambda}_{\leq l}$.

In-between the Hadamard layers, the two sets are coupled via a controlled-phase block, denoted as $\text{CP}_{\Lambda_l, \bar{\Lambda}_{\leq l}}(\theta_{\Lambda_l, \bar{\Lambda}_{\leq l}})$, which applies pairwise operations $\text{CP}_{jk}(\theta_{jk})$ between each qubit $j \in \Lambda_l$ and $k \in \bar{\Lambda}_{\leq l}$. Building upon this notation, we formally define the circuit architecture:

Definition 2 (HP- L circuit). *A quantum circuit is classified as an HP- L circuit if it is constructed using the*

following layer-wise architecture

$$U_{\text{HP}}^{(L)}(\theta) = \bigotimes_{i \in \Lambda_L} H_i \prod_{l=L-1}^1 \left(\text{CP}_{\Lambda_l, \bar{\Lambda}_{\leq l}}(\theta_{\Lambda_l \bar{\Lambda}_{\leq l}}) \bigotimes_{i \in \Lambda_l} H_i \right), \quad (3)$$

where the controlled-phase block is given by

$$\text{CP}_{\Lambda_l, \bar{\Lambda}_{\leq l}}(\theta_{\Lambda_l \bar{\Lambda}_{\leq l}}) = \prod_{j \in \bar{\Lambda}_{\leq l}} \prod_{k \in \Lambda_l} \text{CP}_{jk}(\theta_{jk}). \quad (4)$$

The parameter set $\theta_{\Lambda_l \bar{\Lambda}_{\leq l}}$ is the collection of θ_{jk} , $j \in \Lambda_l$, $k \in \bar{\Lambda}_{\leq l}$.

In the following, we show that HP circuits preserve shift invariance. This implies that such invariance is not unique to the QFT, but is a feature that arises in a wider family of quantum circuits.

Theorem 1. *HP-L circuits are shift-invariant for $\mathbb{Z}_{2^n}$.*

The proof involves giving an analytical expression for the state evolved under an HP-L circuit (Eq.(??)) and checking that shift invariance is indeed respected (Appendix ??).

HP-L for HSP over $\mathbb{Z}$: information on generator is retained.— For HSP over $\mathbb{Z}_{2^n}$, HP-0, namely a parallel Hadamard layer, gives an analytically solvable recovery procedure for the hidden subgroup generator 2^p . The key observation is that the first $n - p$ measurement bits are deterministically zero after applying $H^{\otimes n}$ to a coset state, allowing p to be read off from the zero pattern; the full calculation and sampling algorithm are given in Appendix ???. However, the protocol HP-0 fails for general cyclic groups $\mathbb{Z}_q$ because the measurement distributions corresponding to different subgroups V are not distinguishable. That absence of distinguishability demonstrates that shift invariance alone is insufficient for successful hidden subgroup recovery, and that distinguishability should be imposed as an additional condition.

We use the discrete Fisher information (DFI) as a metric for distinguishability.

$$\text{DFI}(r, n) = \sum_{x=0}^{2^n-1} \frac{(\Pr(x | r+1) - \Pr(x | r))^2}{\Pr(x | r)}, \quad (5)$$

where x represents measurement outcomes, r is the hidden subgroup generator, and $\Pr(x | r)$ is the probability of measuring x when the subgroup generator is r .

For the standard QFT (Appendix ??), we can calculate the DFI as

$$\text{DFI}_{\text{QFT}}(r, n) \approx \frac{4\pi^2}{9} \left(\frac{2^{2n}}{r^2} - 1 \right) \sim \mathcal{O}(2^{2n}). \quad (6)$$

Eq. (??) indicates that the generator r becomes easier to recover as the qubit number increases, which motivates us to limit r to $1 \ll r \ll 2^n$.

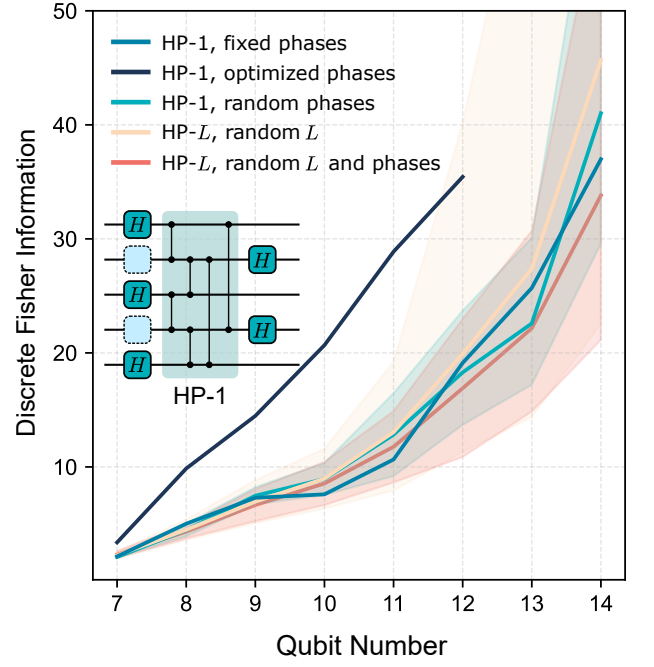


FIG. 2. **Average discrete Fisher information (DFI) increasing with the number of qubits.** Results are shown for several HP-family circuit settings. L denotes the number of phase blocks layers with HP-1 having a single phase block layer. Solid curves show the mean, with shaded bands indicating one standard deviation associated with random phase choices.

For HP circuits, the DFI is evaluated numerically. Specifically, for a fixed r , the circuit maps the periodic input components $\{|qr\rangle\}_q$ to a computational-basis distribution over outcomes $|b\rangle$, with the exact probabilities given by $\Pr(|b\rangle) = \left| \sum_q \langle b | U | qr \rangle \right|^2$. To evaluate the worst-case parameter sensitivity as the system scales, we consider the worst-case DFI: $\text{DFI}_{\min}(n) := \min_{2 \leq r \leq r_{\max}(n)} \text{DFI}(r, n)$, with $r_{\max}(n)^2 \ll 2^n$ in the asymptotic statement. Fig. ?? illustrates the minimum DFI against the qubit number n . (For the finite-size numerical scan, we use the square-root boundary proxy $r_{\max}(n) = \lfloor 2^{n/2} \rfloor$, matching the period scale used in the recovery experiments.)

Theorem 2 (Restricted-window ln-linear DFI bound). *For fixed-phase HP-1, the minimum DFI follows the ln-linear lower bound under $2 < r < 2^{n/4}$,*

$$\text{DFI}_{\min}(n) \gtrsim \exp(kn + b), \quad (7)$$

for constants k and b .

Fig. ?? also compares the scaling of different HP circuit configurations, where solid curves and shaded bands represent the mean and standard deviation of $\text{DFI}_{\min}(n)$. The fixed-phase HP-1 circuit utilizes the skeleton, $\Lambda_1 = \{1, 3, 5, \dots\}$ and $\Lambda_2 = \{2, 4, 6, \dots\}$, with controlled-phase gates with phase $\frac{2\pi}{2^{|i-j|}}$ between qubits i and j .

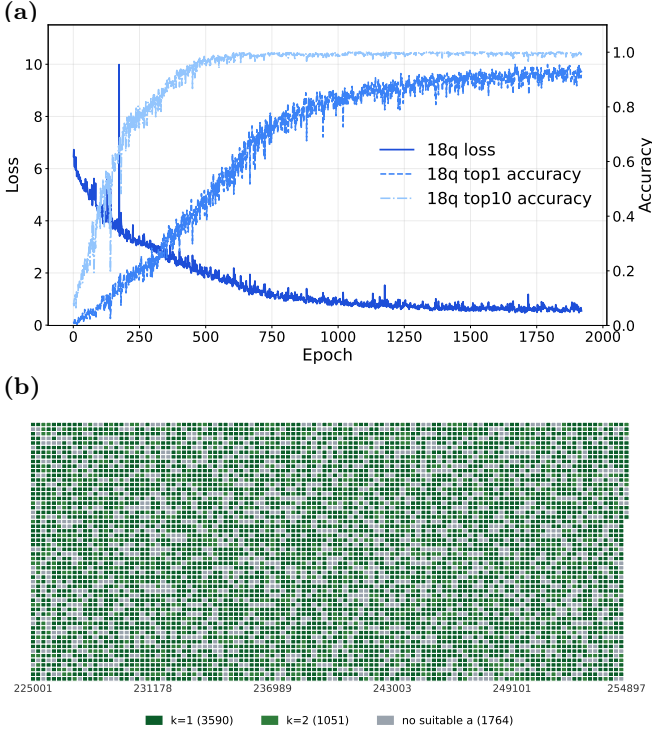


FIG. 3. **Numerical simulation shows successful Shor factoring via HP-1 and neural net post-processing.** (a) Validation loss, top-1 accuracy, and top-10 accuracy for neural-network period recovery on an 18-qubit instance. (b) Shor-type factorization on 18-qubit semiprime instances with $N = 225001, \dots, 254999$. Colour indicates the minimum number k of HP-1-based guesses needed for successful factorization, and grey entries indicate cases where no valid base a yields a detectable period within the chosen bound.

The optimized HP-1 circuit adopts the same skeleton but treats the gate phases as variational parameters, trained to maximize the $\text{DFI}_{\min}$. The deeper HP- L circuits are constructed by generating valid random skeletons Λ .

The proposed HP circuit family preserves an exponential growth trend of DFI over this square-root period window. Appendix ?? reports the corresponding ln-linear regression coefficients.

Efficient neural network for classical post-processing.—To avoid the exponential costs associated with processing full probability distributions, we employ a $\text{poly}(n)$ Deep Sets neural network [?] to predict the hidden subgroup generator directly from the raw measurement outcomes $\{b_i\}$. The input data is generated using the optimized HP-1 circuit evaluated in Fig. ??.

The outcomes are fed into a shared feature map ϕ , which maps each bitstring b_i to a feature vector within 2 linear layers and $16n$ feature dimension. The feature vectors are then summed to form an aggregated, permutation-invariant representation $\sum_i \phi(b_i)$. Subsequently, a beam-search decoder [?] performs a top- k

search to rank candidate periods. We use polynomial samples $m = 1024n^2$ to train the neural net. This architecture avoids exponential sample-space computations and outputs a ranked list of candidate generators for classical verification.

Replacing $\mathcal{O}(n^2)$ QFT with $\mathcal{O}(n)$ HP-1 in Shor’s Algorithm.—We evaluate the optimized HP-1 circuit in classical numerical simulations of Shor’s algorithm for factoring with neural net classical post-processing. Shor’s algorithm seeks the generator r satisfying $a^r \equiv 1 \pmod{N}$. We select the base a such that $1 \leq r \leq 2^{n/2}$, ensuring resolvability by the n -qubit register. We consider factoring 18-qubit semiprime instances with $N = 225001, \dots, 254999$.

The neural net trained successfully. As shown in Fig. ??(a), the neural network achieves rapid convergence when trained on an 18-qubit instance. Both the top-1 (correct period ranked first) and top-10 accuracies increase swiftly within a few epochs.

The combined HP-1 and neural net protocol recovers the subgroup generator in the tested instances. As illustrated in Fig. ??(b), each composite integer is colored by the minimum number of guesses k . Most solved instances are solved on the first attempt ($k = 1$), and the remaining solved instances require $k = 2$.

Fig. ?? reports the 18-qubit gate-level Pauli-noise sweep: top-1 accuracy drops from 0.9059 at 10^{-3} to 0.1412 at $10^{-1.5}$. Details are in Appendix ??.

Summary and outlook.—We propose a hardware-efficient framework for solving the HSP by shallow, shift-invariant HP-1 circuits with a polynomial-scaling neural network. To verify the scalability of this protocol, we analyzed the growth model of the DFI. Finally, we evaluated our protocol on Shor’s algorithm. The numerical simulations demonstrate that our approach successfully finds the hidden subgroup with high accuracy and remains robust against moderate depolarizing noise.

Realizing a practical HSP protocol further requires optimized implementations of the oracle U_f . Related strategies for reducing oracle and data-loading costs include Hamiltonian-learning-based state preparation that transfers the cost to classical preprocessing and uses $\mathcal{O}(1)$ oracle queries [?], space-time tradeoffs for non-Clifford gate reductions [?], and automatic oracle synthesis frameworks [?].

Apart from the $\mathcal{O}(n)$ scaling being hardware friendly, the reduction to grouped phase gates in HP circuits makes the phase operations easier to implement than individual phase gate execution. This hardware-level efficiency arises because consecutive phase operations can be merged into single continuous pulses across multiple experimental platforms, including in linear optics [?], neutral-atom systems [?], trapped ions [?], and NMR [?].

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Code Availability.—All code relevant to this work is publicly available in the GitHub repository: <https://github.com/Fragecity/altqft>.

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- [] D. R. Simon, On the power of quantum computation, *SIAM Journal on Computing* **26**, 1474 (1997).
- [] P. W. Shor, Algorithms for quantum computation: Discrete logarithms and factoring, in *Proceedings 35th Annual Symposium on Foundations of Computer Science* (IEEE, 1994) pp. 124–134.
- [] P. W. Shor, Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer, *SIAM Journal on Computing* **26**, 1484 (1997).
- [] A. Y. Kitaev, Quantum measurements and the abelian stabilizer problem (1995), arXiv:quant-ph/9511026 [quant-ph].
- [] R. Jozsa, Quantum factoring, discrete logarithms, and the hidden subgroup problem, *Computing in Science & Engineering* **3**, 34 (2001).
- [] C. Lomont, The hidden subgroup problem: Review and open problems (2004), arXiv:quant-ph/0411037 [quant-ph].
- [] F. Liu, K. Bian, F. Meng, W. Zhang, and O. Dahlsten, Information compression via hidden subgroup quantum autoencoders, *npj Quantum Information* **10**, 74 (2024).
- [] Y. Kim, A. Eddins, S. Anand, K. X. Wei, E. Van Den Berg, S. Rosenblatt, H. Nayfeh, Y. Wu, M. Zaletel, K. Temme, *et al.*, Evidence for the utility of quantum computing before fault tolerance, *Nature* **618**, 500 (2023).
- [] Google Quantum AI and Collaborators, Quantum error correction below the surface code threshold, *Nature* **638**, 920 (2025).
- [] D. Gao, D. Fan, C. Zha, *et al.*, Establishing a new benchmark in quantum computational advantage with 105-qubit Zuchongzhi 3.0 processor, *Phys. Rev. Lett.* **134**, 090601 (2025).
- [] R. Takagi, S. Endo, S. Minagawa, and M. Gu, Fundamental limits of quantum error mitigation, *npj Quantum Information* **8**, 114 (2022).
- [] R. Takagi, H. Tajima, and M. Gu, Universal sampling lower bounds for quantum error mitigation, *Physical Review Letters* **131**, 210602 (2023).
- [] P. Aumann, M. Fellner, D. Alber, M. Cykier, C. Fleckenstein, R. ter Hoeven, L. Stenzel, R. J. Valencia-Tortora, and W. Lechner, Demonstrating record fidelity for the quantum fourier transform, arXiv preprint arXiv:2604.12465 (2026).
- [] D. Coppersmith, An approximate fourier transform useful in quantum factoring (2002), arXiv:quant-ph/0201067 [quant-ph].
- [] B. Park and D. Ahn, Reducing cnot count in quantum fourier transform for the linear nearest-neighbor architecture, *Scientific Reports* **13**, 8638 (2023).
- [] E. Bäumer *et al.*, Quantum fourier transform using dynamic circuits, arXiv preprint arXiv:2403.09514 (2024).
- [] W. Tang *et al.*, CutQC: using small quantum computers for large quantum circuit evaluations, in *Proceedings of the 26th ACM International Conference on Architectural Support for Programming Languages and Operating Systems* (2021) pp. 473–486.
- [] T. Peng, A. W. Harrow, M. Ozols, and X. Wu, Simulating large quantum circuits on a small quantum computer, *Phys. Rev. Lett.* **125**, 150504 (2020).
- [] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. R. Salakhutdinov, and A. J. Smola, Deep sets, *Advances in neural information processing systems* **30** (2017).
- [] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2000).
- [] H. Edwards and A. Storkey, Towards a neural statistician, arXiv preprint arXiv:1606.02185 (2016).
- [] I. Sutskever, O. Vinyals, and Q. V. Le, Sequence to sequence learning with neural networks, in *Advances in neural information processing systems*, Vol. 27 (2014).
- [] S. Wiseman and A. M. Rush, Sequence-to-sequence learning as beam-search optimization, in *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing* (2016) pp. 1296–1306.
- [] M. Ramezani, S. A. Zargar, S. Salami, A. Bahrampour, and A. Bahrampour, Learning Hamiltonians for $o(1)$ oracle-query quantum state preparation, arXiv preprint arXiv:2512.19181 (2025).
- [] M. Yu, A. T. Calvino, M. Soeken, and G. De Micheli, Back-end-aware fault-tolerant quantum oracle synthesis, in *Proceedings of the 30th Asia and South Pacific Design Automation Conference (ASP-DAC)* (IEEE, 2025) pp. 930–937.
- [] I.-T. Chen and C. Gupta, Towards efficient automatic oracle synthesis and resource estimation using QDK and QIR, in *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)* (IEEE, 2023).
- [] J. Carolan, C. Harrold, C. Sparrow, E. Martín-López, N. J. Russell, J. W. Silverstone, P. J. Shadbolt, N. Matsuda, M. Oguma, M. Itoh, *et al.*, Universal linear optics, *Science* **349**, 711 (2015).
- [] P. Kok, W. J. Munro, K. Nemoto, T. C. Ralph, J. P. Dowling, and G. J. Milburn, Linear optical quantum computing with photonic qubits, *Rev. Mod. Phys.* **79**, 135 (2007).
- [] M. Saffman, T. G. Walker, and K. Mølmer, Quantum information with Rydberg atoms, *Reviews of modern physics* **82**, 2313 (2010).
- [] S. J. Evered, D. Bluvstein, M. Kalinowski, S. Ebadi, T. Manovitz, H. Zhou, S. H. Li, A. A. Geim, T. T. Wang, N. Maskara, *et al.*, High-fidelity parallel entangling gates on a neutral-atom quantum computer, *Nature* **622**, 268 (2023).
- [] M. Mohan, J. De Hond, and S. Kokkelmans, Parametrized multiqubit gates for neutral-atom quantum platforms, *Physical Review Applied* **23**, 054074 (2025).
- [] D. Leibfried, B. DeMarco, V. Meyer, D. Lucas, M. Barrett, J. Britton, W. M. Itano, B. Jelenković, C. Langer, T. Rosenband, and D. J. Wineland, Experimental demonstration of a robust, high-fidelity geometric two ion-qubit phase gate, *Nature* **422**, 412 (2003).
- [] T. R. Tan, J. P. Gaebler, R. Bowler, Y. Lin, J. D. Jost, D. Leibfried, and D. J. Wineland, Demonstration of a

- dressed-state phase gate for trapped ions, Phys. Rev. Lett. **110**, 263002 (2013).
 [] L. M. Vandersypen and I. L. Chuang, NMR techniques for quantum control and computation, Reviews of Mod-

- ern Physics **76**, 1037 (2004).
 [] M. Ettinger, P. Høyer, and E. Knill, The quantum query complexity of the hidden subgroup problem is polynomial, Information Processing Letters **91**, 43 (2004).

APPENDIX

A. Hidden Subgroup Problem and Fourier Sampling

For completeness, this section reviews the standard quantum algorithm for solving the Hidden Subgroup Problem (HSP) over finite abelian groups [? ? ?]. Formally, the HSP is defined as follows:

Definition 3 (Hidden Subgroup Problem). *Given a function $f : G \rightarrow X$ defined on a group G , with the promise that there exists a hidden subgroup $V \leq G$ such that for all $g_1, g_2 \in G$*

$$f(g_1) = f(g_2) \iff g_1 - g_2 \in V \quad (\text{A1})$$

The objective of this task is to identify the hidden subgroup V .

In the context of this paper, we restrict our attention to finite abelian groups. For such groups, the fundamental theorem of finite abelian groups allows us to express the group structure as a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{p_1} \times \mathbb{Z}_{p_2} \times \cdots \times \mathbb{Z}_{p_k}. \quad (\text{A2})$$

The standard quantum procedure for solving the hidden subgroup problem begins by preparing a uniform superposition over all elements of the group G [? ? ?]. We initialize the system in the state

$$|\psi\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle |0\rangle, \quad (\text{A3})$$

where $|G|$ is the number of elements of the group G . The first register holds the group elements, and the second register is initialized to a standard basis state. Next, we evaluate the quantum oracle corresponding to the function f , which acts on the two registers and entangles the group elements with their corresponding function values

$$|\psi\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle |f(g)\rangle. \quad (\text{A4})$$

Grouping the superposition terms according to cosets is illuminating, as the function f is constant within each coset of the hidden subgroup V and distinct across different cosets.

Before proceeding, it is useful to formalize the state notation for a subset. For any subset $S \subseteq G$, we define its corresponding normalized quantum state $|S\rangle$ as the uniform superposition over all elements within the set:

$$|S\rangle = \frac{1}{\sqrt{|S|}} \sum_{s \in S} |s\rangle. \quad (\text{A5})$$

Using this definition, the normalized state for a specific coset $c + V$ is directly given by

$$|c + V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} |c + v\rangle. \quad (\text{A6})$$

By applying this coset state definition, the post-oracle state can be elegantly rewritten as a superposition of these orthogonal coset states, each perfectly entangled with its unique function evaluation:

$$|\psi\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{c \in C} |c + V\rangle |f(c)\rangle, \quad (\text{A7})$$

where C is the set of all possible coset representatives. For simplicity, we assume that the second register is measured immediately after the oracle query. Measuring the second register yields a specific outcome $f(c)$ with a uniform probability of $|V|/|G|$. Consequently, the first register collapses into the corresponding pure coset state:

$$|\psi_{\text{post}}\rangle = |c + V\rangle. \quad (\text{A8})$$

From this point forward, our analysis will focus exclusively on the quantum state of this first register.

Whether the second register is physically measured to collapse the first register into a pure coset state $|c + V\rangle$, or traced out to leave it in a mixed state, the resulting Fourier-basis statistics remain identical [?]. The quantum Fourier transform produces a measurement distribution that is independent of the coset representative. For simplicity, we assume the second register is measured immediately after the oracle query, allowing our subsequent analysis to focus exclusively on the pure coset state:

$$|\psi_{\text{post}}\rangle = |c + V\rangle. \quad (\text{A9})$$

A direct measurement in the computational basis is uninformative. Indeed, using the definition of the coset state,

$$\langle g|c + V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} \langle g|c + v\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} \delta_{g, c+v}. \quad (\text{A10})$$

For fixed c , the map $v \mapsto c + v$ is a bijection from V to the coset $c + V$. Therefore the sum $\sum_{v \in V} \delta_{g, c+v}$ equals 1 when $g \in c + V$ and equals 0 when $g \notin c + V$. Hence

$$\Pr[g | c, V] = |\langle g|c + V\rangle|^2 = \begin{cases} 1/|V|, & g \in c + V, \\ 0, & g \notin c + V, \end{cases} \quad (\text{A11})$$

so the observed element is uniform on the unknown coset. Averaging over uniformly random cosets gives the marginal distribution

$$\Pr[g] = \frac{1}{|G|}, \quad (\text{A12})$$

which contains no direct information about the hidden subgroup V .

To extract meaningful information, we must apply the QFT over the group G before measuring [? ?]. Let $\chi_y(g)$ denote the group characters. The QFT maps a computational basis state $|g\rangle$ as follows:

$$\text{QFT}|g\rangle = \frac{1}{\sqrt{|G|}} \sum_{y \in G} \chi_y(g) |y\rangle. \quad (\text{A13})$$

Applying this transformation to our pure coset state $|c + V\rangle$ yields:

$$\begin{aligned} \text{QFT}|c + V\rangle &= \frac{1}{\sqrt{|V|}} \sum_{h \in V} \text{QFT}|c + h\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{v \in V} \sum_{y \in G} \chi_y(c + v) |y\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{y \in G} \chi_y(c) \left(\sum_{v \in V} \chi_y(v) \right) |y\rangle. \end{aligned} \quad (\text{A14})$$

By the orthogonality of group characters, the sum $\sum_{v \in V} \chi_y(v)$ evaluates to $|V|$ if the character χ_y is trivial on V (i.e., y belongs to the orthogonal subgroup $V^\perp$), and evaluates to 0 otherwise. Thus, the state simplifies to a superposition over $V^\perp$:

$$\text{QFT}|c + V\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{y \in V^\perp} \chi_y(c) |y\rangle. \quad (\text{A15})$$

Measuring this state in the computational basis gives the probability of observing an outcome $y \in V^\perp$:

$$\Pr[y] = \left| \sqrt{\frac{|V|}{|G|}} \chi_y(c) \right|^2 = \frac{|V|}{|G|} |\chi_y(c)|^2 = \frac{|V|}{|G|}. \quad (\text{A16})$$

Crucially, because $|\chi_y(c)|^2 = 1$ for all group characters, the resulting measurement distribution is uniformly distributed over $V^\perp$ and is strictly independent of the random coset representative c . This shift invariance is the

fundamental property that allows the extraction of subgroup information by effectively removing the unknown shift at the probability level.

The QFT over G is defined using the character group $\hat{G}$ by

$$F_G|g\rangle = \frac{1}{\sqrt{|G|}} \sum_{\chi \in \hat{G}} \chi(g) |\chi\rangle, \quad (\text{A17})$$

where $\{|\chi\rangle : \chi \in \hat{G}\}$ is the Fourier basis and each character $\chi : G \rightarrow \mathbb{C}^\times$ is a group homomorphism. Here $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ denotes the multiplicative group of nonzero complex numbers, so with additive notation on G one has $\chi(g_1 + g_2) = \chi(g_1)\chi(g_2)$. For finite abelian G , every character has unit modulus. Applying F_G to a coset state gives

$$\begin{aligned} F_G|c + V\rangle &= \frac{1}{\sqrt{|V|}} \sum_{v \in V} F_G|c + v\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{\chi \in \hat{G}} \sum_{v \in V} \chi(c + v) |\chi\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{\chi \in \hat{G}} \chi(c) \left(\sum_{v \in V} \chi(v) \right) |\chi\rangle. \end{aligned} \quad (\text{A18})$$

The character sum over V satisfies

$$\sum_{v \in V} \chi(v) = \begin{cases} |V|, & \chi \in V^\perp, \\ 0, & \chi \notin V^\perp, \end{cases} \quad V^\perp = \{\chi \in \hat{G} : \chi(v) = 1 \text{ for all } v \in V\}. \quad (\text{A19})$$

Hence

$$F_G|c + V\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{\chi \in V^\perp} \chi(c) |\chi\rangle. \quad (\text{A20})$$

The Fourier-basis measurement distribution is therefore

$$\Pr(\chi \mid c, V) = \begin{cases} |V|/|G|, & \chi \in V^\perp, \\ 0, & \chi \notin V^\perp, \end{cases} \quad (\text{A21})$$

which is independent of the coset representative c . The same formula holds for the mixed state ρ_V , since it is a convex combination of pure coset states with identical Fourier statistics.

Repeating the procedure therefore produces samples from $V^\perp$. In the finite abelian setting, classical post-processing reconstructs $V^\perp$ from the sample span and then recovers the hidden subgroup using the duality identity [? ? ?]

$$(V^\perp)^\perp = V. \quad (\text{A22})$$

Fourier sampling thus converts hidden translational structure in the oracle into explicit linear constraints in the dual group.

B. Necessary condition of shift invariance

Here we make explicit the probability-level shift-invariance condition used in Fourier sampling. Let G be a finite abelian group, and let $\{|g\rangle : g \in G\}$ denote the computational basis. For $c \in G$, define the shift operator

$$T_c|g\rangle = |g + c\rangle. \quad (\text{B1})$$

For a subgroup $V \leq G$, the normalized subgroup state is

$$|c + V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} |c + v\rangle. \quad (\text{B2})$$

For HSP sampling, shift invariance requires that the measurement distribution after applying U is strictly independent of the coset representative:

$$|\langle x|U|V\rangle|^2 = |\langle x|UT_c|V\rangle|^2, \quad \forall x, c \in G, \forall V \leq G. \quad (\text{B3})$$

Operationally, this means $U|V\rangle$ and $U|c+V\rangle$ may differ by a relative phase, but they must yield identical probabilities in the computational basis. Any viable QFT replacement must therefore satisfy this condition.

Proposition (Necessary condition of shift invariance). *Suppose U is a unitary on the group register that satisfies Eq. (??). Then, for every $x, y \in G$,*

$$|\langle x|U|y\rangle| = \frac{1}{\sqrt{|G|}}. \quad (\text{B4})$$

Equivalently, there exist phases $\theta_{xy} \in \mathbb{R}$ such that

$$\langle x|U|y\rangle = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}. \quad (\text{B5})$$

To determine how this restricts the matrix elements of U , we can examine the trivial subgroup $V = \{0\}$. In this case, the subgroup state is simply the basis state $|0\rangle$, meaning the shift-invariance condition directly constrains the columns of U . Since the output probabilities for $|0\rangle$ and any shift $|g\rangle$ must be equal, for any fixed outcome x , all entries in the x -th row of U must share the same modulus. Unitarity then forces this common modulus to be exactly $1/\sqrt{|G|}$. This leads directly to Proposition ??.

Proof of Proposition ??. Take the trivial subgroup $V = \{0\}$. Then $|V\rangle = |0\rangle$ and $T_c|V\rangle = |c\rangle$ for every $c \in G$. Equation (??) therefore implies

$$|\langle x|U|0\rangle|^2 = |\langle x|U|c\rangle|^2, \quad \forall x, c \in G. \quad (\text{B6})$$

Using the matrix element convention $U_{xy} = \langle x|U|y\rangle$, this equality shows that $|U_{x0}|^2 = |U_{xc}|^2$ for all $c \in G$. In other words, for each fixed row x , all entries have the same squared modulus. Denote this common value by a_x . Since U is unitary, the sum of the squared moduli across any row must be normalized to 1:

$$1 = \sum_{y \in G} |U_{xy}|^2 = |G|a_x. \quad (\text{B7})$$

Hence, $a_x = 1/|G|$ for every $x \in G$. Taking the square root yields the uniform modulus for all matrix elements:

$$|U_{xy}| = \frac{1}{\sqrt{|G|}}. \quad (\text{B8})$$

Because this modulus is strictly positive, every entry admits a polar decomposition

$$U_{xy} = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}, \quad \theta_{xy} \in \mathbb{R}. \quad (\text{B9})$$

This proves the claim. $\square$

C. HP- L circuits are shift invariant over $\mathbb{Z}_{2^n}$

This appendix aims to rigorously prove that the HP- L circuit family satisfies the shift-invariance property. Establishing this theorem is of significant practical importance: it fundamentally guarantees that the output statistics obtained from this shallow circuit structure can be reliably used for the classical post-processing reconstruction of the hidden subgroup.

Theorem. *The HP- L circuit is shift-invariant over $\mathbb{Z}_{2^n}$.*

The core strategy of the proof is as follows: First, we explicitly calculate the sequential effect of each layer of the HP- L circuit on the initial basis state, deriving the analytical expression of the final quantum state. Next, we obtain the corresponding probability amplitudes by calculating the matrix elements of this output state in the computational basis. Finally, we compute the squared modulus of these amplitudes over the subgroup V and its coset $c + V$ respectively, using algebraic simplification to factor out a pure phase factor related to the coset representative c . Because the modulus of this pure phase factor is strictly 1, it naturally cancels out when calculating the measurement probabilities, thereby rigorously proving mathematically that the output probability distribution is completely independent of random shifts.

The $\bar{\Lambda}_{\leq l}$ is given by

$$\bar{\Lambda}_{\leq l} = \{1, 2, \dots, n\} \setminus \bigcup_{i \leq l} \Lambda_i. \quad (\text{C1})$$

The $\{\Lambda_l\}$ is a partition of index set $\{1, 2, \dots, n\}$, which means

$$\bigcup \Lambda_l = \{1, 2, \dots, n\}, \quad \Lambda_i \cap \Lambda_j = \emptyset. \quad (\text{C2})$$

Proof. The first step is to calculate the effect of the first layer, explicitly keeping the normalization factor $\frac{1}{\sqrt{2}}$ from the Hadamard gates:

$$\begin{aligned} & \text{CP}_{\Lambda_1, \bar{\Lambda}_{\leq 1}}(\theta_{\Lambda_1 \bar{\Lambda}_{\leq 1}}) \bigotimes_{i \in \Lambda_1} H_i |b_1 b_2 \dots b_n\rangle \\ &= \text{CP}_{\Lambda_1, \bar{\Lambda}_{\leq 1}}(\theta_{\Lambda_1 \bar{\Lambda}_{\leq 1}}) \bigotimes_{i \in \Lambda_1} \frac{1}{\sqrt{2}} (|0\rangle + e^{\pi i b_i} |1\rangle) \bigotimes_{k \in \bar{\Lambda}_1} |b_k\rangle \end{aligned} \quad (\text{C3})$$

$$= \bigotimes_{i \in \Lambda_1} \frac{1}{\sqrt{2}} \left(|0\rangle + \exp \left[\pi i b_i + i \sum_{j \in \bar{\Lambda}_{\leq 1}} \theta_{ij} b_j \right] |1\rangle \right) \bigotimes_{k \in \bar{\Lambda}_1} |b_k\rangle. \quad (\text{C4})$$

Note that since the qubits in $\bar{\Lambda}_{\leq 1}$ have not yet been acted upon by Hadamard gates, they remain in the computational basis.

Since the subsets Λ_l form a partition of the qubits:

$$\bigcup_{l=1}^L \Lambda_l = \{1, 2, \dots, n\}, \quad \Lambda_i \cap \Lambda_j = \emptyset, \quad (\text{C5})$$

and the subsets $\bar{\Lambda}_l$ are defined as the remaining qubits that have not yet been acted on by Hadamard gates:

$$\bar{\Lambda}_{\leq l} = \{1, 2, \dots, n\} \setminus \left(\bigcup_{k=1}^l \Lambda_k \right), \quad (\text{C6})$$

the sequential action of all L layers ensures every qubit i is acted on by a Hadamard gate exactly once. Let $l(i)$ denote the specific stage where $i \in \Lambda_{l(i)}$. The final state after the full circuit becomes:

$$U_{\text{HP}}^{(L)}(\theta) |b_1 b_2 \dots b_n\rangle = \bigotimes_{i=1}^n \frac{1}{\sqrt{2}} \left(|0\rangle + \exp \left[\pi i b_i + i \sum_{j \in \bar{\Lambda}_{l(i)}} \theta_{ij} b_j \right] |1\rangle \right). \quad (\text{C7})$$

To find the matrix element, we take the inner product with $\langle b'_1 b'_2 \dots b'_n |$. For each qubit i , the $|1\rangle$ component is selected if and only if $b'_i = 1$, which mathematically corresponds to multiplying the entire phase exponent by b'_i . Thus, we obtain:

$$\begin{aligned} & \langle b'_1 b'_2 \dots b'_n | U_{\text{HP}}^{(L)}(\theta) |b_1 b_2 \dots b_n\rangle \\ &= \frac{1}{\sqrt{2}^n} \prod_{i=1}^n \exp \left[\pi i b_i b'_i + i \sum_{j \in \bar{\Lambda}_{l(i)}} \theta_{ij} b'_i b_j \right] \end{aligned} \quad (\text{C8})$$

$$= \frac{1}{\sqrt{2^n}} \exp \left[i\pi \sum_{i=1}^n b_i b'_i + i \sum_{l=1}^{L-1} \sum_{i \in \Lambda_l} \sum_{j \in \bar{\Lambda}_{\leq l}} \theta_{ij} b'_i b_j \right]. \quad (\text{C9})$$

To simplify the notation for the matrix elements while respecting the sequential asymmetry of the circuit (where each qubit pair interacts only once), we define a directed effective phase matrix $\tilde{\Theta}$ whose elements are given by:

$$\tilde{\theta}_{ij} = \begin{cases} \theta_{ij}, & \text{if } i \in \Lambda_{l(i)}, j \in \bar{\Lambda}_{l(i)} \text{ (i.e., } l(i) < l(j)) \\ \pi, & \text{if } i = j \\ 0, & \text{otherwise (i.e., } l(i) \geq l(j)). \end{cases} \quad (\text{C10})$$

Using this effective matrix, the exponent can be compactly written as a full summation over all i, j :

$$\langle b'_1 b'_2 \cdots b'_n | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | b_1 b_2 \cdots b_n \rangle = \frac{1}{\sqrt{2^n}} \exp \left[i \sum_{i,j} \tilde{\theta}_{ij} b'_i b_j \right]. \quad (\text{C11})$$

The shift invariance requires that the probability distribution over the subgroup V is identical to its cosets. First, we compute the squared probability amplitude over V :

$$\left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | V \rangle \right|^2 = \frac{1}{2^n |V|} \left| \sum_{y \in V} \exp \left[i \sum_{i,j} \tilde{\theta}_{ij} x_i y_j \right] \right|^2. \quad (\text{C12})$$

Since $V \leq \mathbb{Z}_{2^n}$, $V = \{q2^s \mid q \in \mathbb{Z}\}$ for certain $s \leq n$. Assuming a standard binary encoding where the s least significant bits correspond to the trailing qubits, $y \in V$ can be denoted as $y = y_1 y_2 \cdots y_{n-s} 0 \cdots 0$, with $y_i \in \{0, 1\}$. Similarly, the coset $c + V = \{q2^s + c \mid 0 \leq c < 2^s, q \in \mathbb{Z}\}$, and its elements $y \in c + V$ can be denoted by matching the trailing bits to the binary representation of c , yielding $y = y_1 y_2 \cdots y_{n-s} c_{n-s+1} \cdots c_n$, with $y_i, c_k \in \{0, 1\}$.

Therefore, we can expand the squared amplitude for the subgroup V :

$$\begin{aligned} & \left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | V \rangle \right|^2 \\ &= \frac{1}{2^n |V|} \left| \sum_{y \in V} \exp \left[i \sum_{i=1}^n \sum_{j=1}^n \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \end{aligned} \quad (\text{C13})$$

$$= \frac{1}{2^n 2^{n-s}} \left| \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2. \quad (\text{C14})$$

Evaluating the same expression for the coset $c + V$, we factor out the terms involving the constant bits c_k :

$$\begin{aligned} & \left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | c + V \rangle \right|^2 \\ &= \frac{1}{2^n |c + V|} \left| \sum_{y \in c + V} \exp \left[i \sum_{i=1}^n \sum_{j=1}^n \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \end{aligned} \quad (\text{C15})$$

$$= \frac{1}{2^n 2^{n-s}} \left| \exp \left[i \sum_{i=1}^n \sum_{k=n-s+1}^n \tilde{\theta}_{ik} x_i c_k \right] \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \quad (\text{C16})$$

$$= \frac{1}{2^n 2^{n-s}} \left| \exp \left[i \sum_{i=1}^n \sum_{k=n-s+1}^n \tilde{\theta}_{ik} x_i c_k \right] \right|^2 \left| \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2. \quad (\text{C17})$$

Because the modulus of the extracted pure phase factor strictly equals 1, it completely cancels out, yielding:

$$\left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | c + V \rangle \right|^2 = \frac{1}{2^n 2^{n-s}} \left| \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \quad (\text{C18})$$

$$= \left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | V \rangle \right|^2. \quad (\text{C19})$$

This completes the proof. $\square$

D. Sampling calculation and recovery guarantee for the HP – 0 case

We now give the calculation underlying the HP-0 discussion in the main text. Consider $G \simeq \mathbb{Z}_{2^n}$ represented by n computational-basis qubits. Because $\mathbb{Z}_{2^n}$ is a cyclic group of order 2^n , any of its subgroups must be generated by a divisor of 2^n . Thus, the hidden subgroup is necessarily of the form $V = \langle 2^p \rangle$ for some integer $0 \leq p \leq n$. Since every element $h \in V$ is a multiple of 2^p , its n -bit binary representation naturally ends with p zeros. Under this standard convention, an element h takes the form

$$h = h_1 h_2 \cdots h_{n-p} 0 \cdots 0. \quad (\text{D1})$$

The subgroup generated by 2^p is

$$V = \{q2^p : q = 0, \dots, 2^{n-p} - 1\}, \quad (\text{D2})$$

so $|V| = 2^{n-p}$. Since adding an element of V can vary the first $n-p$ bits, each coset has a representative of the form

$$c = 0 \cdots 0 c_{n-p+1} \cdots c_n. \quad (\text{D3})$$

Thus a normalized coset state can be written as

$$|c + V\rangle = \frac{1}{\sqrt{2^{n-p}}} \sum_{h_1, \dots, h_{n-p} \in \{0,1\}} |h_1 \cdots h_{n-p} c_{n-p+1} \cdots c_n\rangle. \quad (\text{D4})$$

For an n -bit string x , applying Hadamard gates to all n qubits gives

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} (-1)^{k \cdot x} |k\rangle, \quad (\text{D5})$$

where $k \cdot x = \sum_{i=1}^n k_i x_i$ denotes the standard bitwise inner product of the binary strings k and x . Applying this to Eq. (??) gives

$$\begin{aligned} H^{\otimes n} |c + V\rangle &= \frac{1}{\sqrt{2^{n-p}}} \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} \sum_{h_1, \dots, h_{n-p} \in \{0,1\}} (-1)^{\sum_{i=1}^{n-p} k_i h_i + \sum_{i=n-p+1}^n k_i c_i} |k\rangle \\ &= \frac{1}{\sqrt{2^{n-p}}} \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} (-1)^{\sum_{i=n-p+1}^n k_i c_i} \prod_{i=1}^{n-p} \left(\sum_{h_i \in \{0,1\}} (-1)^{k_i h_i} \right) |k\rangle. \end{aligned} \quad (\text{D6})$$

The inner product factor is zero unless $k_1 = \cdots = k_{n-p} = 0$, in which case it equals 2^{n-p} . Therefore

$$H^{\otimes n} |c + V\rangle = \frac{1}{\sqrt{2^p}} \sum_{k_{n-p+1}, \dots, k_n \in \{0,1\}} (-1)^{\sum_{i=n-p+1}^n k_i c_i} |0\rangle^{\otimes (n-p)} |k_{n-p+1} \cdots k_n\rangle. \quad (\text{D7})$$

The coset representative c is eliminated in Eq. (??). Consequently, the measurement distribution is independent of c . The first $n-p$ bits are always zero, while the remaining p bits are uniformly random.

This immediately gives a simple reconstruction procedure for the idealized $\mathbb{Z}_{2^n}$ HSP. Each application of the oracle-and-measurement routine prepares a random coset state, and Eq. (??) shows that subsequently applying $H^{\otimes n}$ converts that coset state into a sample whose first $n-p$ bits are deterministically zero while the remaining p bits are unbiased. The resulting sampling-and-recovery routine is summarized in Algorithm ??.

We note that the physical measurement of the second register in Algorithm ?? is not strictly necessary. By the principle of deferred measurement, tracing out the second register leaves the first register in a mixed state of all possible cosets. Since the measurement distribution is strictly independent of the coset representative c , this yields

Algorithm 1: HP – 0 Sampling for $V \leq \mathbb{Z}_{2^n}$

Input: Oracle access to an HSP instance over $\mathbb{Z}_{2^n}$, sample count m
Output: Hidden subgroup $V = \langle 2^p \rangle$

```

1 Initialize an empty list of samples  $\mathcal{B}$ ;
2 for  $i = 1$  to  $m$  do
3   Prepare the uniform superposition  $\frac{1}{\sqrt{2^n}} \sum_{g \in \mathbb{Z}_{2^n}} |g\rangle |0\rangle$ ;
4   Query the oracle (measuring the second register to obtain a coset state  $|c_i + V\rangle$  is optional);
5   Apply  $H^{\otimes n}$  to the first register;
6   Measure the first register and record the bit string  $b^{(i)} \in \{0, 1\}^n$ ;
7   Append  $b^{(i)}$  to  $\mathcal{B}$ ;
8 end
9 Find the minimum index  $j^*$  (where  $j = 1$  is the most significant bit) such that  $b_{j^*}^{(i)} = 1$  for some sample in  $\mathcal{B}$ ;
10 if such a  $j^*$  exists then
11   Let  $p = n - j^* + 1$ ;
12 else
13   Let  $p = 0$ ;
14 end
15 return  $V = \langle 2^p \rangle$ ;
```

identical measurement statistics. This device-friendly feature avoids the need for mid-circuit measurements, which is highly advantageous for implementations on near-term quantum hardware.

Algorithm ?? is correct because, under the bit-ordering convention above, the free positions occur exactly in the final p bits. Equivalently, the measurement data have the form

$$0 \cdots 0 * \cdots *, \quad (\text{D8})$$

where each star denotes an unbiased bit. Therefore, the classical post-processing only needs to identify the leftmost bit position that ever evaluates to 1 across all m samples. This single position j^* uniquely determines p , and hence determines the subgroup $V = \langle 2^p \rangle$.

The sample complexity is logarithmic in the number of qubits. Under this leftmost-1 decoding strategy, the algorithm only fails if the most significant free bit (at position $n - p + 1$) is measured as 0 in all m independent samples. Because this specific bit is uniformly random, the probability of it being zero m times in a row is exactly 2^{-m} . Thus $m \geq \lceil \log_2(1/\epsilon) \rceil$ samples are sufficient for Algorithm ?? to perfectly identify the hidden subgroup V with probability at least $1 - \epsilon$. This tight bound elegantly removes the union-bound dependence on n , yielding an optimized sample complexity.

E. Fisher information for subgroup-generator estimation

To benchmark the proposed circuit family, we treat the hidden subgroup generator r as the parameter to be inferred from the measurement distribution produced by the standard QFT. Consider the truncated periodic state

$$R := \left\lfloor \frac{2^n - 1}{r} \right\rfloor + 1, \quad (\text{E1})$$

where R is the number of supported multiples of r , and

$$|\psi_r\rangle := \frac{1}{\sqrt{R}} \sum_{q=0}^{R-1} |qr\rangle. \quad (\text{E2})$$

Thus $q = 0, \dots, R-1$ labels precisely the integers qr contained in the 2^n -dimensional register. For the 2^n -point QFT,

$$|j\rangle \mapsto \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i j k / 2^n} |k\rangle, \quad (\text{E3})$$

and therefore

$$|\psi_r\rangle \mapsto \frac{1}{\sqrt{R 2^n}} \sum_{k=0}^{2^n-1} \left(\sum_{q=0}^{R-1} e^{2\pi i q r k / 2^n} \right) |k\rangle. \quad (\text{E4})$$

The corresponding measurement probability is

$$\Pr(k|r) = \frac{1}{R2^n} \left| \sum_{q=0}^{R-1} e^{2\pi i q r k / 2^n} \right|^2 = \frac{1}{R2^n} \left(\frac{\sin(\pi R r k / 2^n)}{\sin(\pi r k / 2^n)} \right)^2. \quad (\text{E5})$$

In the continuum approximation, the Fisher-information calculation reduces to a Dirichlet-kernel moment estimate. The result needed for the main-text comparison is summarized in the following lemma.

Lemma 1 (QFT Fisher-information). *In the regime $1 \ll r \ll 2^n$, the QFT Fisher information is approximated by*

$$\text{DFI}_{\text{QFT}}(r, n) \approx \frac{4\pi^2}{9} \left(\frac{2^{2n}}{r^2} - 1 \right). \quad (\text{E6})$$

Proof. In the regime $r \ll 2^n$, introduce the continuum measurement coordinate

$$x := \frac{\pi k}{2^n} \in [0, \pi]. \quad (\text{E7})$$

The corresponding continuum approximation to the QFT probability density is

$$f(x, r) := \frac{1}{\pi R} \left(\frac{\sin(Rrx)}{\sin(rx)} \right)^2. \quad (\text{E8})$$

The QFT Fisher information with respect to the generator r is

$$\text{DFI}_{\text{QFT}}(r, n) = \int_0^\pi f(x, r) \left[\frac{\partial}{\partial r} \ln f(x, r) \right]^2 dx, \quad (\text{E9})$$

with score function

$$s_r(x) := \frac{\partial}{\partial r} \ln f(x, r) = 2Rx \cot(Rrx) - 2x \cot(rx), \quad s_r(x) = \text{score for the generator } r. \quad (\text{E10})$$

Define

$$D_R(u) := \frac{\sin(Ru)}{\sin u}, \quad D_R(u) = \text{order-}R \text{ Dirichlet kernel}. \quad (\text{E11})$$

Then $f(x, r) = D_R(rx)^2 / (\pi R)$, and the score function implies

$$f(x, r) \left[\frac{\partial}{\partial r} \ln f(x, r) \right]^2 = \frac{4x^2}{\pi R} [D'_R(rx)]^2. \quad (\text{E12})$$

Thus the Fisher-information integral is reduced to the square of a finite trigonometric polynomial. Once that polynomial is expanded into cosine modes, the remaining calculation involves only the moments of x^2 against those modes.

We begin by expressing the Dirichlet kernel $D_R(u)$ in its exponential sum form. By shifting the summation index to center the phases, it can be written as a symmetric finite series:

$$D_R(u) = \frac{\sin(Ru)}{\sin u} = \sum_{m=0}^{R-1} e^{i(R-1-2m)u}, \quad (\text{E13})$$

Differentiating this series term-by-term with respect to u yields:

$$D'_R(u) = i \sum_{m=0}^{R-1} (R-1-2m) e^{i(R-1-2m)u}. \quad (\text{E14})$$

To compute the squared magnitude $[D'_R(u)]^2$, we multiply this sum by itself. By collecting terms that produce the same relative frequency difference, we can perfectly recast the product as an even cosine series:

$$[D'_R(u)]^2 = A_0 + 2 \sum_{j=1}^{R-1} A_j \cos(2ju), \quad (\text{E15})$$

where A_j denotes the cosine-series coefficient at frequency $2j$. These coefficients are obtained by evaluating the convolution sum over amplitudes with frequency offset j . To evaluate this sum explicitly, we expand the summand as a quadratic in m and apply the standard summation formulas for $\sum m$ and $\sum m^2$:

$$\begin{aligned}
A_j &= - \sum_{m=0}^{R-1-j} (R-1-2m)(2j-R+1+2m) \\
&= \sum_{m=0}^{R-1-j} [4m^2 - 4(R-1-j)m + (R-1-j)^2 - j^2] \\
&= \frac{R-j}{3} [(R-j-1)^2 + 2(R-j-1) - 3j^2] \\
&= \frac{(R-j)(R^2 - 2Rj - 2j^2 - 1)}{3}, \quad j = 0, \dots, R-1.
\end{aligned} \tag{E16}$$

In particular, setting $j = 0$ isolates the non-oscillatory constant term:

$$A_0 = \frac{R(R^2 - 1)}{3}. \tag{E17}$$

Recall from our earlier derivation of the score function that the Fisher information integral takes the form:

$$\text{DFI}_{\text{QFT}}(r, n) = \frac{4}{\pi R} \int_0^\pi x^2 [D'_R(rx)]^2 dx, \tag{E18}$$

Substituting our exact cosine expansion of $[D'_R(rx)]^2$ into this integral distributes the integration across the individual frequency modes:

$$\text{DFI}_{\text{QFT}}(r, n) = \frac{4}{\pi R} \left[A_0 \int_0^\pi x^2 dx + 2 \sum_{j=1}^{R-1} A_j \int_0^\pi x^2 \cos(2jrx) dx \right]. \tag{E19}$$

The problem is now reduced to evaluating two elementary definite integrals over $[0, \pi]$. Define the non-oscillatory and oscillatory moments by

$$M_0 := \int_0^\pi x^2 dx = \frac{\pi^3}{3}, \tag{E20}$$

$$\begin{aligned}
M_j(r) &:= \int_0^\pi x^2 \cos(2jrx) dx \\
&= \frac{4\pi jr \cos(2\pi jr) + (4\pi^2 j^2 r^2 - 2) \sin(2\pi jr)}{(2jr)^3}, \quad j \geq 1.
\end{aligned} \tag{E21}$$

Here M_0 is the non-oscillatory second moment, and $M_j(r)$ is the oscillatory second moment at mode j . Inserting these evaluated moments, along with the explicit expressions for A_0 and A_j , back into the expanded Fisher information equation gives:

$$\begin{aligned}
\text{DFI}_{\text{QFT}}(r, n) &= \frac{4}{\pi R} \left[\frac{R(R^2 - 1)}{3} \cdot \frac{\pi^3}{3} + \frac{2}{3} \sum_{j=1}^{R-1} (R-j)(R^2 - 2Rj - 2j^2 - 1) M_j(r) \right] \\
&= \frac{4\pi^2}{9} (R^2 - 1) + \frac{8}{3\pi R} \sum_{j=1}^{R-1} (R-j)(R^2 - 2Rj - 2j^2 - 1) M_j(r),
\end{aligned} \tag{E22}$$

which is the stated exact formula.

Finally, we analyze the asymptotic behavior in the standard regime of interest, $1 \ll r \ll R$. In this limit, the hidden parameter is large enough to suppress oscillations, yet small enough that its multiples are heavily supported in the register. The explicit form of the oscillatory moments shows that $M_j(r) = \mathcal{O}(r^{-2})$ for any fixed mode j . Since the leading constant term scales as $\mathcal{O}(R^2)$ and the summation over the modes scales proportionally to $\mathcal{O}(R^2 r^{-2})$,

the relative magnitude of the oscillatory corrections is heavily suppressed by a factor of $1/r^2$. Because these high-frequency corrections are strictly subleading in this large- r regime, the behavior is overwhelmingly governed by the constant contribution. Keeping only this non-oscillatory term and substituting $R \approx 2^n/r$ therefore gives:

$$\text{DFI}_{\text{QFT}}(r, n) \approx \frac{4\pi^2}{9} \left(\frac{2^{2n}}{r^2} - 1 \right). \quad (\text{E23})$$

This proves the lemma. $\square$

Lemma 2 (HP-1 period-law overlap). *For the fixed-phase HP-1 circuit, let $\Pr(\cdot | s)$ and $\Pr(\cdot | t)$ denote the coherent output probability laws associated with two candidate periods s, t . Let $\nu_2(m)$ denote the largest integer a such that 2^a divides m , and set $a = \nu_2(\gcd(s, t))$. Then, for a constant κ_a depending on a , their overlap obeys the following statistical law:*

$$\sum_x \Pr(x | s) \Pr(x | t) \approx \frac{1}{2^n} + \kappa_a \frac{2^a}{st}, \quad a = \nu_2(\gcd(s, t)). \quad (\text{E24})$$

Proof. Set $N := 2^n$. The goal is to evaluate the period-law overlap by expanding both coherent probability laws and then carrying out the sum over output strings x . This separates the overlap into a diagonal contribution, two single-period off-diagonal contributions, and one genuinely mixed off-diagonal contribution. For the fixed-phase HP-1 circuit, let $C = C_{\text{HP1}} \in \mathbb{R}^{n \times n}$ denote the HP-1 phase matrix normalized by

$$\langle x | U_{\text{HP}}^{(1)} | y \rangle = \frac{1}{\sqrt{N}} \exp(\pi i x^T C y), \quad x, y \in \{0, 1\}^n. \quad (\text{E25})$$

Equivalently, C is the directed fixed-phase HP-1 matrix with nonzero entries

$$C_{ij} = \begin{cases} \frac{1}{2^{|i-j|}}, & i = j \text{ or } (i \in \Lambda_1, j \in \Lambda_2), \\ 0, & \text{otherwise,} \end{cases} \quad (\text{E26})$$

For a candidate period s , define

$$R_s := |\{q \in \mathbb{Z}_{\geq 0} : 0 \leq qs < N\}| = \left\lfloor \frac{N-1}{s} \right\rfloor + 1. \quad (\text{E27})$$

Let $\mathbf{b}_m \in \{0, 1\}^n$ denote the n -bit binary representation of the integer m . Using Eq. (??), the probability law $\Pr(\cdot | s)$ is

$$\begin{aligned} \Pr(x | s) &= \left| \frac{1}{\sqrt{R_s}} \sum_{q=0}^{R_s-1} \langle x | U_{\text{HP}}^{(1)} | \mathbf{b}_{qs} \rangle \right|^2 \\ &= \frac{1}{NR_s} \left| \sum_{q=0}^{R_s-1} \exp(\pi i x^T C \mathbf{b}_{qs}) \right|^2, \quad x \in \{0, 1\}^n. \end{aligned} \quad (\text{E28})$$

Substituting Eq. (??) and the corresponding expression for $\Pr(\cdot | t)$ into the overlap on the left-hand side of Eq. (??) gives

$$\begin{aligned} \sum_x \Pr(x | s) \Pr(x | t) &= \frac{1}{N^2 R_s R_t} \sum_{x \in \{0, 1\}^n} \left| \sum_{q=0}^{R_s-1} \exp(\pi i x^T C \mathbf{b}_{qs}) \right|^2 \\ &\quad \times \left| \sum_{\ell=0}^{R_t-1} \exp(\pi i x^T C \mathbf{b}_{\ell t}) \right|^2. \end{aligned} \quad (\text{E29})$$

We next perform the x -sum exactly. Since x ranges over all binary strings, the sum factorizes bit by bit. The following kernel records the real contribution of each conjugate pair that appears after expanding the two squared moduli. For $v \in \mathbb{Z}^n$, the associated conjugate-pair identity defines the real cosine kernel $\Omega(v)$ by

$$\prod_{a=1}^n \left[1 + e^{\pi i (Cv)_a} \right] + \prod_{a=1}^n \left[1 + e^{-\pi i (Cv)_a} \right] = 2N \cos \left[\frac{\pi}{2} \sum_{a=1}^n (Cv)_a \right] \prod_{a=1}^n \cos \left[\frac{\pi}{2} (Cv)_a \right] =: 2N\Omega(v). \quad (\text{E30})$$

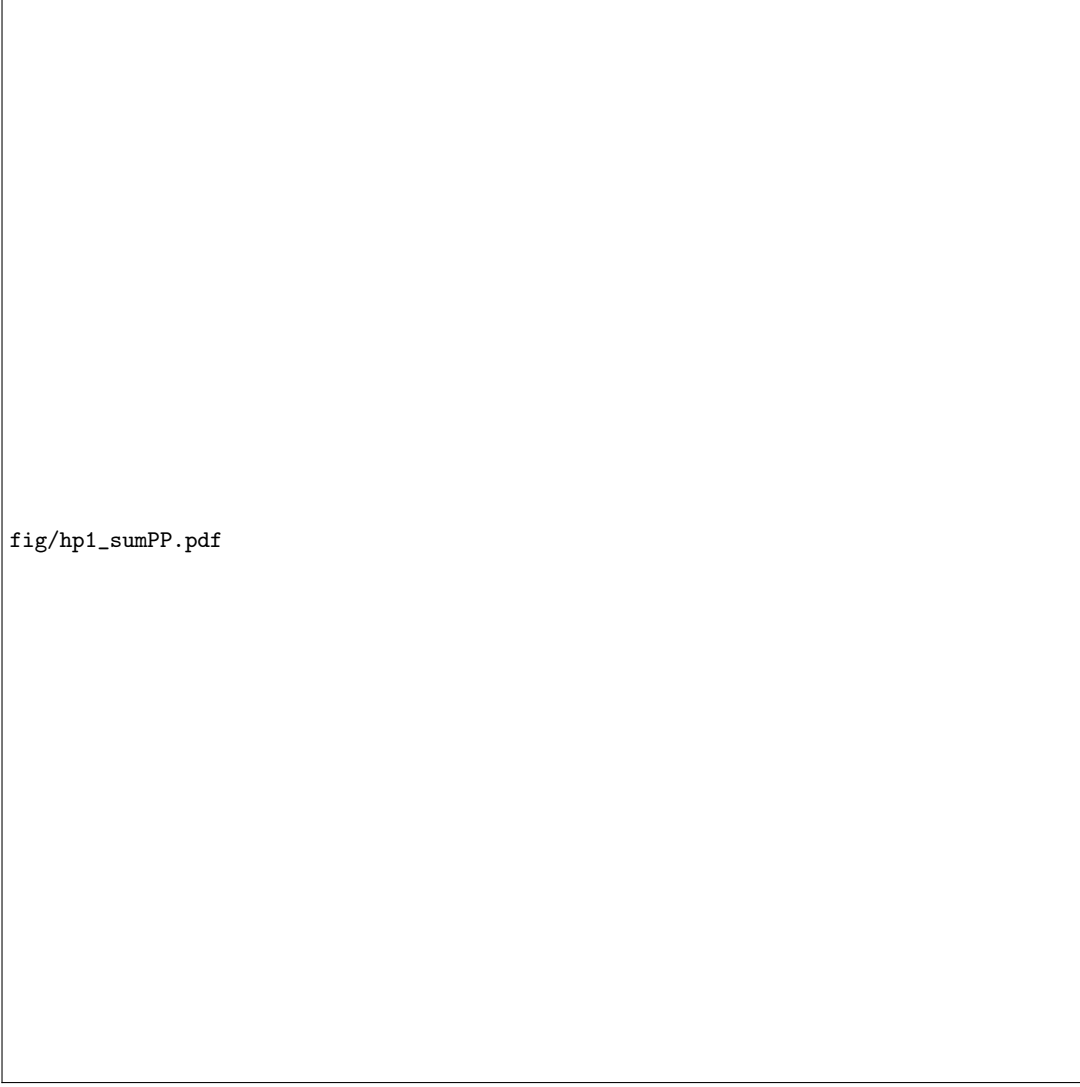


FIG. 4. Conditioned Monte Carlo estimate of $B_{s,t}^{(n)}$ for fixed-phase HP-1 at $n = 200$. For each $a = 0, \dots, 150$, 100 dyadic pairs are sampled. The dashed curve shows Eq. (??).

For fixed q, q', ℓ, ℓ' , set

$$\Delta_{q,q',\ell,\ell'} := \mathbf{b}_{qs} - \mathbf{b}_{q's} + \mathbf{b}_{\ell t} - \mathbf{b}_{\ell't}.$$

Expanding the two squared moduli in Eq. (??), exchanging the order of summation, and substituting the x -sum kernel explicitly gives

$$\begin{aligned} \sum_x \Pr(x | s) \Pr(x | t) &= \frac{1}{N^2 R_s R_t} \sum_{q,q'=0}^{R_s-1} \sum_{\ell,\ell'=0}^{R_t-1} \prod_{a=1}^n [1 + \exp(\pi i (C \Delta_{q,q',\ell,\ell'})_a)] \\ &= \left[\frac{1}{N} + \frac{2}{N R_s} \sum_{0 \leq q < q' < R_s} \Omega(\Delta_{q,q',0,0}) + \frac{2}{N R_t} \sum_{0 \leq \ell < \ell' < R_t} \Omega(\Delta_{0,0,\ell,\ell'}) \right. \\ &\quad \left. + \frac{2}{N R_s R_t} \sum_{0 \leq q < q' < R_s} \sum_{0 \leq \ell < \ell' < R_t} [\Omega(\Delta_{q,q',\ell,\ell'}) + \Omega(\Delta_{q,q',\ell',\ell})] \right]. \end{aligned} \quad (\text{E31})$$

Here $\Delta_{q,q',0,0} = \mathbf{b}_{qs} - \mathbf{b}_{q's}$ and $\Delta_{0,0,\ell,\ell'} = \mathbf{b}_{\ell t} - \mathbf{b}_{\ell't}$, since the paired diagonal difference is zero. The first equality is the fully substituted form of the x -sum: no additional kernel notation is being used. The passage from the first line

of Eq. (??) to the second line comes from grouping ordered indices into the smallest real orbits. The conjugate-pair identity in Eq. (??) is applied to each conjugate pair. If $q = q'$ and $\ell = \ell'$, then $\Delta = 0$, the product equals N , and the $R_s R_t$ such terms give the first normalized contribution $1/N$ inside the brackets. If $q \neq q'$ while $\ell = \ell'$, then for each unordered pair $q < q'$ and each of the R_t diagonal values of ℓ , the two ordered terms (q, q') and (q', q) have opposite vectors $\pm(\mathbf{b}_{qs} - \mathbf{b}_{q's})$, giving the normalized coefficient $2/(NR_s)$. The case $q = q'$, $\ell \neq \ell'$ is identical and gives the coefficient $2/(NR_t)$. Finally, when both ordered pairs are off diagonal, the four terms generated by $(q, q') \leftrightarrow (q', q)$ and $(\ell, \ell') \leftrightarrow (\ell', \ell)$ split into the two conjugate pairs $\pm\Delta_{q,q',\ell,\ell'}$ and $\pm\Delta_{q,q',\ell',\ell}$, giving the final double sum with normalized coefficient $2/(NR_s R_t)$. Thus the remaining sums run over unordered indices $q < q'$ and $\ell < \ell'$. Individual Ω -terms need not be nonnegative; the nonnegativity of the whole expression follows from its equality to the probability overlap $\sum_x \Pr(x | s) \Pr(x | t)$.

The two single-period off-diagonal sums in Eq. (??) are in fact killed by a zero factor in the cosine product. For $q \neq q'$, set

$$u = \mathbf{b}_{qs} - \mathbf{b}_{q's}. \quad (\text{E32})$$

Since $u \neq 0$ and each component of u belongs to $\{-1, 0, 1\}$, some component u_a is odd. If a changed component lies in Λ_2 , then row a of C is diagonal and gives the zero factor $\cos(\pi u_a/2) = 0$. If no component in Λ_2 changes, then some component in Λ_1 changes; all Λ_2 -tails vanish in that row, so again $(Cu)_a = u_a = \pm 1$ and the same zero factor appears. The same argument with $q = q'$ gives the corresponding single- t cancellation:

$$\begin{aligned} \Omega(\Delta_{q,q',0,0}) &= 0, & q \neq q', \\ \Omega(\Delta_{0,0,\ell,\ell'}) &= 0, & \ell \neq \ell'. \end{aligned} \quad (\text{E33})$$

The mixed term is different because two such bit-difference vectors are added. Writing

$$\Delta_{q,q',\ell,\ell'} = u + w, \quad w = \mathbf{b}_{\ell t} - \mathbf{b}_{\ell' t}, \quad (\text{E34})$$

one has

$$u_a, w_a \in \{-1, 0, 1\}, \quad (\Delta_{q,q',\ell,\ell'})_a \in \{-2, -1, 0, 1, 2\}. \quad (\text{E35})$$

For the mixed off-diagonal contribution, we define the rescaled quantity

$$B_{s,t}^{(n)} := \frac{1}{2^{n-1} R_s R_t} \sum_{0 \leq q < q' < R_s} \sum_{0 \leq \ell < \ell' < R_t} [\Omega(\Delta_{q,q',\ell,\ell'}) + \Omega(\Delta_{q,q',\ell',\ell})]. \quad (\text{E36})$$

This is the only term in Eq. (??) that is not forced to vanish by the single-period cancellation. The normalization by $2^{n-1} R_s R_t$ removes the explicit system-size factor from the mixed double sum, so that different n and different dyadic scales can be compared on the same plot. Numerically, we observe a simple statistical law for this mixed contribution. For dyadic periods $s = 2^a u$ and $t = 2^a v$, with u, v odd and $a = \nu_2(\gcd(s, t))$, the conditioned Monte Carlo estimate in Fig. ?? is well described by

$$B_{s,t}^{(n)} \approx \kappa_a \frac{2^a}{st}. \quad (\text{E37})$$

Thus Fig. ?? supplies the empirical closure for the remaining mixed double sum: after the above normalization, its leading dependence is $2^a/(st)$, up to the coefficient κ_a . At the level of bit differences, an odd change in one period difference can be cancelled by the other difference, or two equal changes can combine into an even component, for example $1 + (-1) = 0$, $1 + 1 = 2$, and $(-1) + (-1) = -2$. These cancellations can remove the zero cosine factor, so the mixed Ω -terms are not identically zero, even though they are numerically sparse for random off-diagonal indices.

Finally, substituting the single-period cancellation Eq. (??) and the statistical law Eq. (??) into Eq. (??), together with the definition of $B_{s,t}^{(n)}$ in Eq. (??), gives

$$\sum_x \Pr(x | s) \Pr(x | t) \approx \frac{1}{2^n} + \kappa_a \frac{2^a}{st}, \quad a = \nu_2(\gcd(s, t)). \quad (\text{E38})$$

This is the overlap formula stated in Eq. (??). □

Theorem ?? (Restricted-window ln-linear DFI bound). *For fixed-phase HP-1, the minimum DFI follows the ln-linear lower bound under $2 < r < 2^{n/4}$,*

$$\text{DFI}_{\min}(n) \gtrsim \exp(kn + b), \quad (\text{E39})$$

for constants k and b .

Proof of Theorem ??. We prove the claim by converting the discrete Fisher information into a collision-overlap estimate for two neighboring period distributions. The quantity in the theorem compares the sampling laws at periods r and $r + 1$. Thus the first step is to isolate the part of the DFI that measures how far these two laws are from each other, while keeping track of the normalizing denominator coming from the law at r :

$$\text{DFI}(r, n) = \sum_{x \in \{0,1\}^n} \frac{(\Pr(x | r + 1) - \Pr(x | r))^2}{\Pr(x | r)}. \quad (\text{E40})$$

The denominator in Eq. (??) is inconvenient for direct use of Lemma ??, which controls quadratic overlaps of probability laws. We therefore introduce exactly the two quadratic objects needed for the reduction: the overlap between two period laws and the squared L^2 distance between adjacent laws,

$$S_{s,t} := \sum_{x \in \{0,1\}^n} \Pr(x | s) \Pr(x | t), \quad (\text{E41})$$

$$D_r := \sum_{x \in \{0,1\}^n} (\Pr(x | r + 1) - \Pr(x | r))^2. \quad (\text{E42})$$

This definition makes the numerator algebraic. Expanding the square in D_r produces one self-overlap for period $r + 1$, one self-overlap for period r , and one mixed overlap. Hence

$$D_r = S_{r+1,r+1} - 2S_{r+1,r} + S_{r,r}. \quad (\text{E43})$$

We next remove the pointwise denominator in the DFI. The self-overlap $S_{r,r}$ is the collision probability of the period- r output law, so it controls every point mass of that law. Since each summand in $S_{r,r}$ is nonnegative, for every x we have

$$\Pr(x | r)^2 \leq \sum_y \Pr(y | r)^2 = S_{r,r}, \quad \Pr(x | r) \leq \sqrt{S_{r,r}}. \quad (\text{E44})$$

Consequently each nonzero denominator in Eq. (??) is at most $\sqrt{S_{r,r}}$. This converts the DFI into a lower bound depending only on the adjacent L^2 distance, and then on the three overlaps above:

$$\begin{aligned} \text{DFI}(r, n) &= \sum_x \frac{(\Pr(x | r + 1) - \Pr(x | r))^2}{\Pr(x | r)} \\ &\geq \frac{1}{\sqrt{S_{r,r}}} \sum_x (\Pr(x | r + 1) - \Pr(x | r))^2 \\ &= \frac{D_r}{\sqrt{S_{r,r}}} = \frac{S_{r+1,r+1} - 2S_{r+1,r} + S_{r,r}}{\sqrt{S_{r,r}}}. \end{aligned} \quad (\text{E45})$$

At this point the problem has been reduced to estimating the three overlap terms in Eq. (??). The diagonal terms measure the concentration of each individual period law, while the mixed term measures how much the two adjacent period laws overlap. These are precisely the quantities provided by Lemma ?. Applying that lemma to the pair $r, r + 1$ gives

$$\begin{aligned} S_{r,r} &\approx \frac{1}{2^n} + \kappa_{\nu_2(r)} \frac{2^{\nu_2(r)}}{r^2}, \\ S_{r+1,r+1} &\approx \frac{1}{2^n} + \kappa_{\nu_2(r+1)} \frac{2^{\nu_2(r+1)}}{(r+1)^2}, \\ S_{r+1,r} &\approx \frac{1}{2^n} + \frac{\kappa_0}{r(r+1)}, \end{aligned} \quad (\text{E46})$$

where $\nu_2(\gcd(r, r+1)) = 0$. The common 2^{-n} term is the uniform background contribution. It appears in all three overlaps, so it cancels in the numerator $S_{r+1, r+1} - 2S_{r+1, r} + S_{r, r}$. Substituting Eq. (??) into Eq. (??) therefore gives the following explicit lower bound:

$$\text{DFI}(r, n) \gtrsim \frac{\kappa_{\nu_2(r+1)} \frac{2^{\nu_2(r+1)}}{(r+1)^2} - \frac{2\kappa_0}{r(r+1)} + \kappa_{\nu_2(r)} \frac{2^{\nu_2(r)}}{r^2}}{\sqrt{\frac{1}{2^n} + \kappa_{\nu_2(r)} \frac{2^{\nu_2(r)}}{r^2}}}. \quad (\text{E47})$$

We now identify the leading behavior of the right-hand side. For the generic adjacent case, assume $\{\nu_2(r), \nu_2(r+1)\} = \{0, 1\}$. This means one of the two adjacent periods carries one factor of 2, while the other is odd. In either parity ordering, the leading r^{-2} contributions in the numerator combine in the same way. Thus the numerator of Eq. (??) satisfies, in either parity case,

$$\kappa_{\nu_2(r+1)} \frac{2^{\nu_2(r+1)}}{(r+1)^2} - \frac{2\kappa_0}{r(r+1)} + \kappa_{\nu_2(r)} \frac{2^{\nu_2(r)}}{r^2} = \frac{2\kappa_1 - \kappa_0 + \mathcal{O}(r^{-1})}{r^2}. \quad (\text{E48})$$

The denominator is a collision scale for the period- r law. In the window used for the theorem, $r^2 \ll 2^n$, so the structured collision term $\kappa_{\nu_2(r)} 2^{\nu_2(r)} / r^2$ dominates the uniform background 2^{-n} . Hence the denominator has size $1/r$:

$$\sqrt{\frac{1}{2^n} + \kappa_{\nu_2(r)} \frac{2^{\nu_2(r)}}{r^2}} = \frac{\sqrt{c_r + o(1)}}{r}, \quad c_r := \kappa_{\nu_2(r)} 2^{\nu_2(r)} \in \{\kappa_0, 2\kappa_1\}, \quad (\text{E49})$$

where $r^2 \ll 2^n$ was used. Combining the r^{-2} numerator scale with the r^{-1} denominator scale leaves a $1/r$ lower bound. Therefore, for $2\kappa_1 > \kappa_0$,

$$\text{DFI}(r, n) \gtrsim \frac{2\kappa_1 - \kappa_0 + \mathcal{O}(r^{-1})}{\sqrt{c_r + o(1)}} \frac{1}{r} = \Omega\left(\frac{1}{r}\right). \quad (\text{E50})$$

This is the main estimate: within the generic adjacent class, the DFI cannot decay faster than a reciprocal period scale. The theorem then follows by taking the minimum over the restricted window. Since $r < 2^{n/4}$, the reciprocal-period lower bound has an exponential dependence on n on the natural-log scale. Absorbing the fixed prefactors and window-endpoint effects into the constants gives the stated ln-linear exponential form

$$\text{DFI}_{\min}(n) := \min_{2 \leq r < 2^{n/4}} \text{DFI}(r, n) \gtrsim \exp(kn + b), \quad (\text{E51})$$

for constants k and b . □

F. Log-linear growth fit for the numerical Fisher-information

The fixed- r QFT benchmark derived above clarifies the dominant scaling mechanism, but Fig. ?? probes a slightly different quantity. There, the vertical axis is the minimum discrete Fisher information over the period window used in the numerical scan. To ensure the asymptotic condition $1 \ll r \ll 2^n$ is strictly maintained as the system scales, the numerical scan uses the polynomial period window $2 \leq r \leq n^2$. The relevant QFT comparator is therefore the worst case within this dynamically scaled window rather than a single fixed value of r .

The leading term in Lemma ?? is monotone in R , and in the sampled regime $R \approx 2^n/r$. Consequently, the smallest QFT Fisher information inside the scan window is attained near the largest sampled period. Substituting $r_{\max}(n) = n^2$ yields the window-matched estimate

$$\text{DFI}_{\text{QFT}, \min}(n) \approx \frac{4\pi^2}{9} \left(\left\lfloor \frac{2^n}{n^2} \right\rfloor^2 - 1 \right). \quad (\text{F1})$$

If one keeps only the dominant exponential factor, the logarithmic growth becomes

$$\ln \text{DFI}_{\text{QFT}, \min}(n) = 2n \ln 2 - 4 \ln n + \mathcal{O}(1). \quad (\text{F2})$$

Thus, the asymptotic QFT exponent under this polynomial period scaling approaches $k_{\text{QFT},\infty} = 2\ln 2$, with a finite-window reduction from the $-4\ln n$ term.

This observation suggests a simple diagnostic for the numerical data in Fig. ???. If the plotted curves retain a stable exponential dependence on the qubit count over the sampled window, then the transformed data should be well approximated by

$$\ln \text{DFI}_{\min}(n) = kn + b. \quad (\text{F3})$$

We therefore perform ordinary least-squares fits to the log-transformed mean values of each curve shown in Fig. ???. Table ?? summarizes the fitted exponents; we do not repeat the transformed plot because it carries the same point set as Fig. ??.

Circuit	qubits	k	95% CI	R^2
HP-1 fixed	7–18	0.368	[0.324, 0.413]	0.972
HP-1 shared optimized	7–18	0.379	[0.354, 0.403]	0.992
HP- L random	7–18	0.398	[0.383, 0.414]	0.997
HP- L random with phase	7–18	0.352	[0.336, 0.367]	0.996

TABLE I. Effective exponents obtained from the log-linear model $\ln \text{DFI}_{\min}(n) = kn + b$ applied to the numerical Fisher-information curves in Fig. ??.

The fitted lines account for most of the log-scale variance across the sampled window. The coefficient of determination (R^2) ranges from 0.972 to 0.997, the log-space root-mean-square residual lies between 0.075 and 0.217, and every fitted slope (k) is significantly positive with a p -value $\leq 4.6 \times 10^{-9}$. In that sense, a linear regression on $\ln I_{\min}$ is justified as a finite-window descriptive model for these curves. At the same time, the fitted slope k should be interpreted as an effective exponent on the restricted range $n = 7, \dots, 18$ (where n is the number of qubits), not as a claim about the ultimate asymptotic law.

The comparison with the QFT benchmark is unambiguous. Even after matching the period window used in the numerical scan, the QFT reference fit remains much steeper, with $k_{\text{QFT},\text{win}} = 1.089$ and $R^2 = 0.999$. The empirical circuit families therefore grow substantially more slowly: their fitted exponents occupy 32%–37% of the window-matched QFT slope, and 25%–29% of the asymptotic QFT exponent $2\ln 2$. Among the plotted families, the random-layer HP circuit has the largest point estimate, near $k = 0.398$, followed by the shared optimized HP-1 circuit, near $k = 0.379$.

This comparison sharpens the interpretation of Fig. ???. The HP family circuits retain a clear exponential-in- n growth trend in the $\text{DFI}_{\min}$ over the sampled window, which is why the log-linear fit works well, but they do not reproduce the QFT growth rate. Their advantage in the present setting is therefore not that they match the full QFT benchmark asymptotically, but that they retain a nontrivial and practically usable fraction of that information while remaining much shallower.

G. Decoder Model

The measurement distribution produced by an HP circuit depends on the hidden period. For example, Fig. ?? shows the output distributions for two different periods, $r = 3$ and $r = 5$. The locations and weights of the prominent peaks change with r , demonstrating that the samples retain information about the hidden generator. Unlike the standard QFT distribution, however, these peaks do not exhibit a simple number-theoretic pattern that can be directly converted into a continued-fraction estimate. The irregular structure makes a compact analytical decoder difficult to construct.

An ideal generic decoder would be the maximum-likelihood estimator (MLE). Let $B_m = \{b_i\}_{i=1}^m$ denote m independent measurement outcomes, with $b_i \in \{0, 1\}^n$, and let $\Pr(b \mid r)$ be the output probability of the HP circuit for a candidate period r . The MLE is

$$\hat{r}_{\text{ML}} = \arg \max_{r \in \mathcal{R}} \prod_{i=1}^m \Pr(b_i \mid r) = \arg \max_{r \in \mathcal{R}} \sum_{i=1}^m \log \Pr(b_i \mid r), \quad (\text{G1})$$

where $\mathcal{R}$ is the candidate-period set. Under the usual regularity conditions for likelihood estimation, the Fisher information determines the local asymptotic precision of the estimator, so its inverse sets a local sample-precision

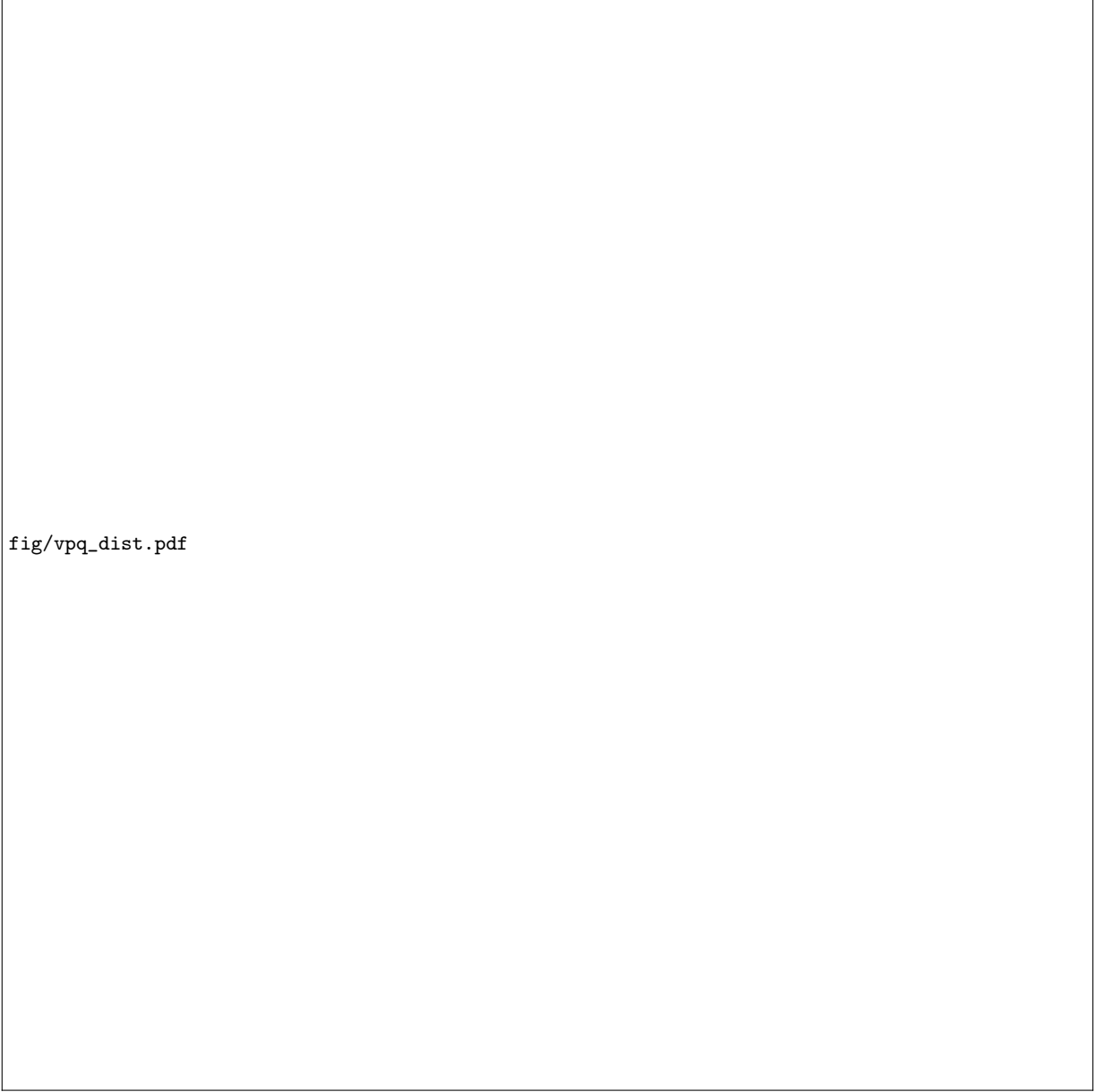


FIG. 5. **Period-dependent HP output distributions.** Output probabilities for two hidden periods, $r = 3$ and $r = 5$. The period-dependent peak structure shows that the measurement samples retain information about r , while the irregular peak locations do not provide an obvious closed-form decoding rule.

scale [? ?]. For the discrete period problem, the global error exponent additionally depends on the separation between the full distributions $\Pr(\cdot | r)$ for different r ; therefore the DFI alone should not be interpreted as a proof of an achievable decoding error exponent. The empirical MLE failure rates shown in Fig. ?? decrease rapidly as the number of samples increases, and are approximately exponential over the displayed range. This observation provides a useful numerical benchmark for the amount of information available in the HP samples, but it is not an asymptotic theorem for all periods or circuit families.

The MLE is not the decoder used in the main numerical protocol. Evaluating Eq. (??) requires computing $\Pr(b_i | r)$ for every observed bit string and every candidate period. A direct construction of the complete output distribution contains 2^n probabilities and requires repeated circuit simulation, which is prohibitive in the regime of interest. A specialized likelihood evaluation with lower cost is possible in principle, but its implementation and complexity are not established here. We therefore replace the explicit likelihood calculation by a learned estimator that acts directly on the sampled bit strings.

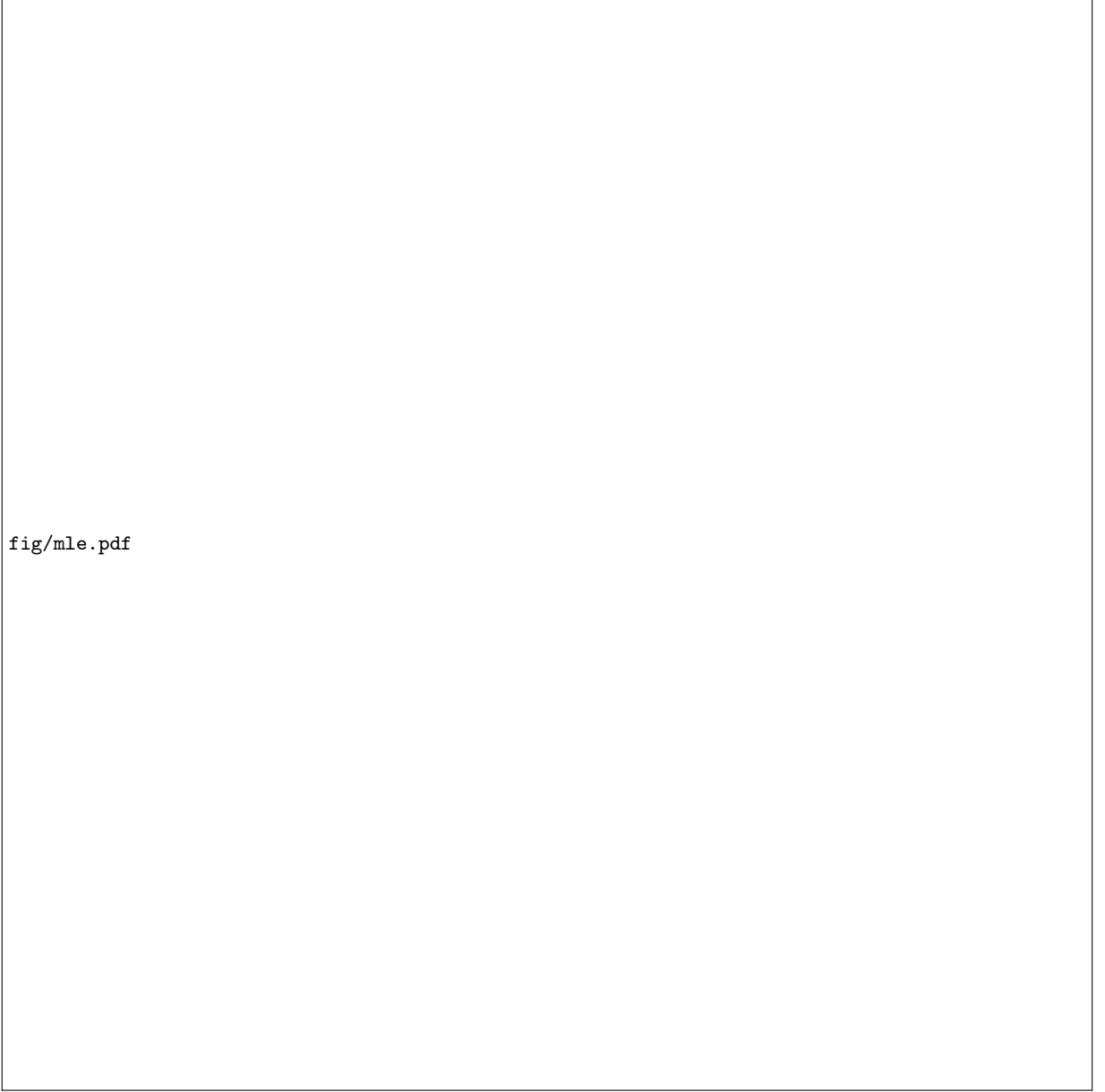


FIG. 6. **Maximum-likelihood decoding benchmark.** Empirical failure rate of the MLE as a function of the number of measurement samples for $n = 8, 9, 10$, and 11 . The rapid decrease in failure rate illustrates the improvement obtained from additional samples in the tested regime.

The decoder used in the 18-qubit experiments is a weighted, permutation-invariant Deep Sets model [? ?]. Let $d_e = \max(8, n)$ and $d = 16n$. For bit position j , the measured value b_{ij} and its position have independently learned embeddings $E_{\text{val}}(b_{ij}), E_{\text{pos}}(j) \in \mathbb{R}^{d_e}$. Their sum is concatenated over the n positions, layer-normalized, and projected from nd_e to d dimensions, followed by layer normalization and a GELU activation. This representation is processed by three residual MLP blocks. Each block has the pre-normalized form

$$\mathcal{B}(h) = h + D[W_2 D(\text{GELU}(W_1 \text{LN}(h) + c_1)) + c_2], \quad W_1 : \mathbb{R}^d \rightarrow \mathbb{R}^{2d}, \quad W_2 : \mathbb{R}^{2d} \rightarrow \mathbb{R}^d, \quad (\text{G2})$$

where D denotes dropout. The resulting shared map is denoted by $\phi : \{0, 1\}^n \rightarrow \mathbb{R}^d$.

Repeated outcomes are stored once together with their counts. If $\{\tilde{b}_u\}_{u=1}^U$ are the distinct observed strings and c_u

are their multiplicities, the invariant representation is the count-weighted mean

$$z(B_m) = \sum_{u=1}^U w_u \phi(\tilde{b}_u), \quad w_u = \frac{c_u}{\sum_{v=1}^U c_v}. \quad (\text{G3})$$

This is exactly the ordinary mean $m^{-1} \sum_i \phi(b_i)$ without explicitly repeating identical strings; zero-weight padding permits batching sets with different numbers of distinct outcomes. After another layer normalization, two more residual blocks of the form in Eq. (??) act on $z(B_m)$. A final linear layer produces one logit for each candidate period. In the reported model $n = 18$, so $d_e = 18$, $nd_e = 324$, $d = 288$, and the hidden dimension in each residual block is $2d = 576$. The output classes are the 510 periods $\mathcal{R} = \{2, \dots, 511\}$, and the complete network has 1,909,230 trainable parameters.

The implemented scoring head is a multiclass classifier rather than an autoregressive sequence model. Applying log-softmax to its 510 logits and sorting them gives the exact top- k candidates; the experiments retain $k = 10$ for subsequent classical verification. Thus, the beam label in the schematic architecture refers only to retaining a short ranked list: no stepwise beam expansion or separately chosen beam width is used. At inference, the protocol supplies $m = 1024n^2$ circuit measurements (331,776 for $n = 18$); count compression and Eq. (??) make the network input depend on the number of distinct outcomes rather than on storing all m strings separately.

For training, cached HP-1 output distributions were constructed for every $r \in \{2, \dots, 511\}$ and every subgroup shift $s \in \{0, \dots, r-1\}$. One shift per period was reserved before training, leaving 130,305 period-shift rows available for training. Each data item was generated by drawing 8192 outcomes from one cached distribution and compressing them to distinct bit strings with empirical-frequency weights. At each epoch, 96 periods were selected and 1024 training items were drawn; the mini-batch size was 8. Validation used a fixed subset of 512 draws from the held-out shifts.

The network was trained with AdamW using learning rate 10^{-3} , weight decay 10^{-4} , the default momentum parameters $(\beta_1, \beta_2) = (0.9, 0.999)$, and cross-entropy loss with label smoothing 0.02. All residual-block dropout probabilities were 0.1. The random seed was 7, and CUDA training used bfloat16 autocasting. The 510-class run was initialized from a checkpoint trained on 322 candidate periods: all shape-compatible shared tensors and the classifier rows corresponding to overlapping periods were copied. Checkpoints were selected by validation top-1 accuracy, with a minimum of 300 epochs and early-stopping patience 250. Training stopped after epoch 1919; the selected checkpoint was epoch 1669, with validation loss 0.4845, top-1 accuracy 0.9551, and top-10 accuracy 1.0000.

The use of a neural decoder should be distinguished from an analytical decoding theorem. The DFI characterizes the sensitivity of the HP output distribution to the hidden period and is therefore used here as an information-theoretic resolution scale [?]. Once the likelihood is replaced by a trained neural estimator, the DFI is not an asymptotically achieved bound for that estimator; the actual decoder performance must be established empirically through validation accuracy, top- k recovery, and the factorization tests. In this sense, the neural network demonstrates that the information diagnosed by the DFI is practically accessible in the tested regime.

H. Noise-robustness experiment

We test noise robustness on the trained 18-qubit HP-1 neural-decoder pipeline used for the period-recovery experiments. Both the learned circuit phases and the neural-network weights are kept fixed throughout the sweep. The candidate periods are $r \in \{2, \dots, 511\}$, giving 510 period-recovery instances, and the representative subgroup shift is fixed to $s = 0$. The decoder is the weighted Deep Sets model selected at epoch 1669, whose noiseless validation accuracy is 0.9551 for top-1 recovery and 1.0000 for top-10 recovery. For every period and noise strength, the decoder input is a batch of $1024n^2 = 331776$ sampled measurement outcomes with $n = 18$.

The noise model is gate-level stochastic Pauli depolarization, not a post-circuit mixture with the uniform distribution. We rebuild the optimized 18-qubit Qiskit circuit from the learned shared HP-1 phase parameters; the circuit contains 18 Hadamard gates and 51 controlled-phase gates. After each Hadamard gate, an error is sampled independently: with probability η , one of X, Y, Z is applied uniformly. After each controlled-phase gate, a two-qubit error is sampled independently: with probability η , one of the 15 non-identity two-qubit Pauli products is applied uniformly. The noisy measurement distributions are then passed through the trained decoder by forward pass only, with no retraining, calibration, or noise-aware fine tuning.

The sweep uses 20 geometrically spaced noise strengths in

$$\eta \in [10^{-3}, 10^{-1.5}]. \quad (\text{H1})$$

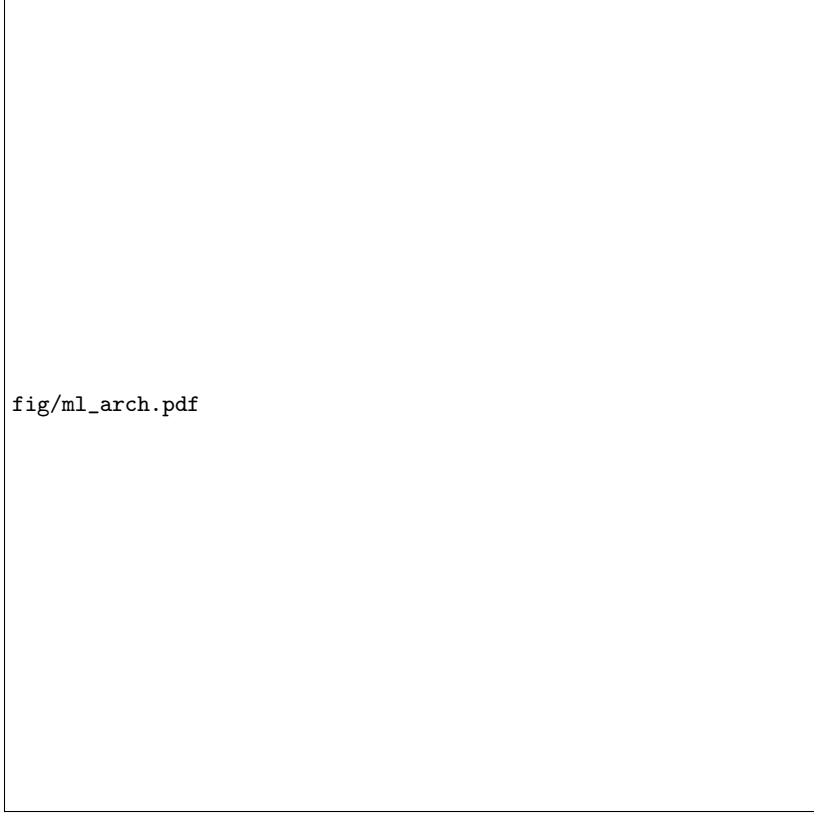


FIG. 7. **Neural decoder architecture.** The HP circuit produces an unordered set of measurement bit strings. A shared feature map processes every bit string, the resulting features are aggregated into a permutation-invariant representation, and beam search produces a ranked list of candidate periods.

For each η , we average four noisy-circuit Monte Carlo trajectories per period before resampling the 331776 measurement outcomes. The resulting top-1 accuracy is plotted in Fig. ?? . Accuracy starts at 0.9059 for $\eta = 10^{-3}$, remains above 0.84 through $\eta = 2.98 \times 10^{-3}$, decreases to 0.7118 at $\eta = 6.16 \times 10^{-3}$, and reaches 0.1412 at $\eta = 10^{-1.5}$. This monotonic degradation over the tested window is consistent with accumulated gate errors and contrasts with the much weaker effect of applying only a single post-circuit global depolarizing mixture.

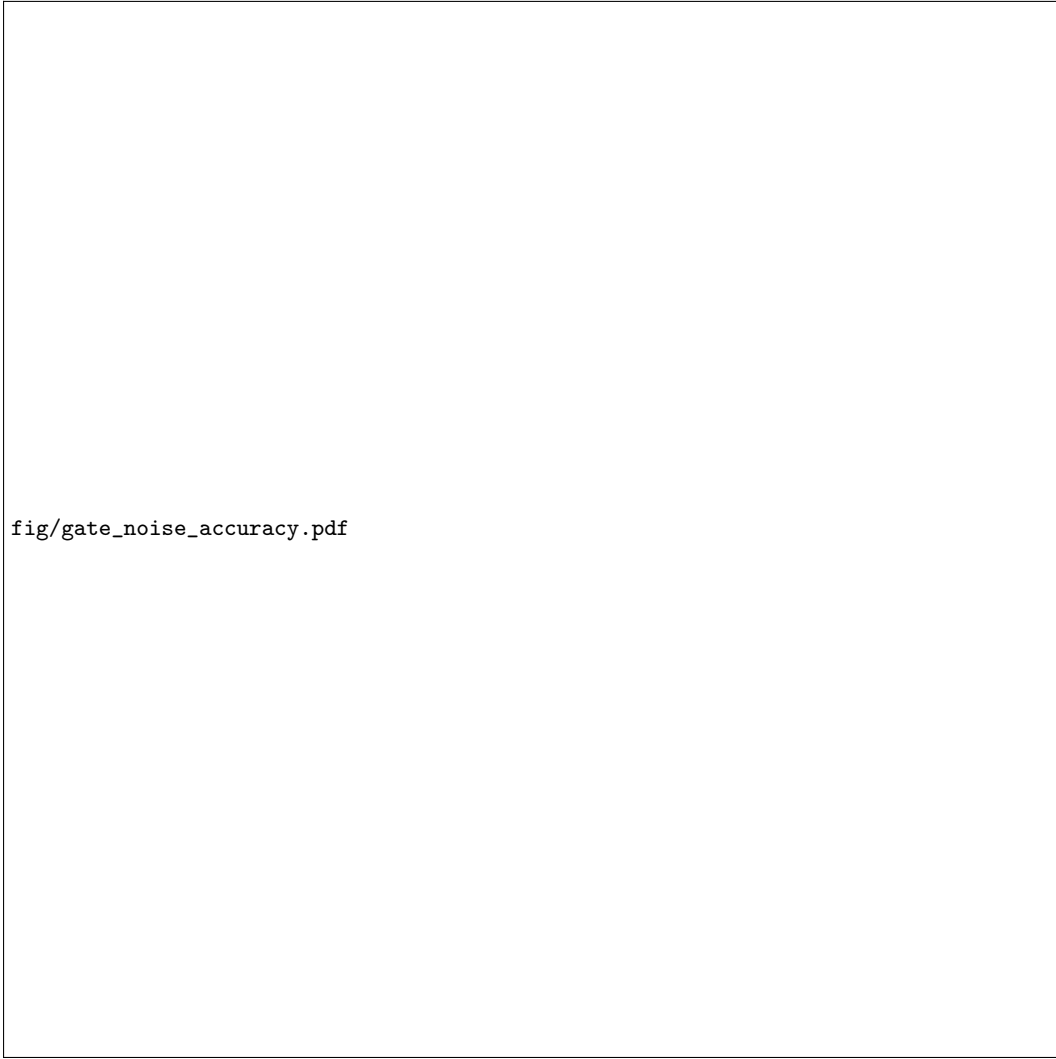


FIG. 8. **Gate-level noise robustness.** Top-1 period-recovery accuracy of the fixed 18-qubit HP-1 neural decoder under stochastic Pauli errors inserted after every Hadamard and controlled-phase gate. The sweep evaluates 510 candidate periods, averages four noisy-circuit trajectories per period, and uses $331776 = 1024n^2$ samples at $n = 18$.